SCOB032 Computer Networks

Deep Study Guide

Mobile IP and wireless security · Network layer · Application layer · Transport layer · Data link and medium access · Physical layer and communication channels

Built from the lecture decks (slides embedded as screenshots) and Tanenbaum 5e / Forouzan 5e. University of Limpopo, Dr R.L. Moila.

Part	Topic	Lecture deck
A	Mobile IP (from vocabulary), wireless security, DoS, encryption, defences	MOBILE NETWORKS, slides 33 to 63
B	Network layer: IP, routing, congestion, QoS, internetworking, subnetting worked examples	Network Layer + subnetting exercises
C	Application layer: client and server side, HTTP, streaming, content delivery	Application Layer, from slide 22
D	Transport layer: ports, TCP/UDP, handshake math, congestion and flow control	Chapter 5 Transport Layer
E	Data link and medium access: CSMA/CD, CSMA/CA, ALOHA, error control	Datalink layer + Medium Access Protocols
F	Physical layer: impairments, guided/unguided media, multiplexing, switching	Communication Channels + Physical layer
G	Logic map and pre-exam checklist	all

How to use this guide

This guide is built from Dr Moila's lecture decks (the slides are embedded as screenshots next to the notes that explain them) and from Tanenbaum 5e and Forouzan 5e where the slides are thin. Where the sources differ, **one explanation is chosen** (the one best suited to the exam) and used throughout. Every concept ends with a small **↳ link** line that shows where it connects, so you can follow the logic flow of the whole course.

Tag	Meaning
[T1]	Term Test 1 scope: OSI troubleshooting, topologies, media, switching, parity/Hamming
[T2]	Term Test 2 / Special Test 2 scope: VLSM, TCP handshake, ports, wireless/mobile security
[D1]–[D5]	Final exam domains: D1 Physical, D2 Data Link/MAC, D3 Network/Routing, D4 Transport, D5 Application/Security

The logic flow of the course (one stack, bottom to top)

Layer	What it does	Part of this guide
Physical	Moves raw bits: media, noise, multiplexing, switching	**Part F**
Data link	Frames, errors, who may transmit (MAC)	**Part E**
Network	Addresses, routing, subnets, Mobile IP	**Part B** and **Part A**
Transport	End-to-end delivery: ports, TCP/UDP, handshake	**Part D**
Application	DNS, DHCP, web, email, streaming, CDN	**Part C**
Security	CIA, attacks, encryption, defences (cuts across all)	**Part A (second half)**

> Read the parts in order A to F only if you like top-down. For first-time learning, read **F → E → B → D → C → A**: each layer builds on the one below it.

PART A: Mobile IP and wireless security

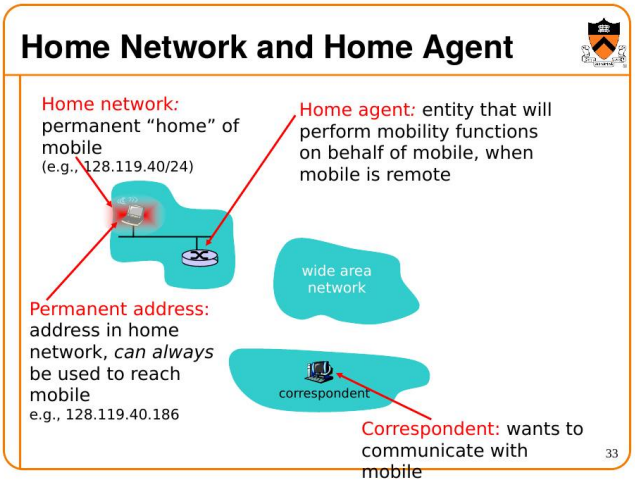
Source: MOBILE NETWORKS deck, slides 33 to 63 (starting at the Mobile IP vocabulary). Textbooks: Tanenbaum §5.6.9 and Ch. 8; Forouzan Ch. 19 and Ch. 31 to 32. **Exam tags:** [D3], [D5], [T2].

A.1 Why Mobile IP exists (one-minute recap)

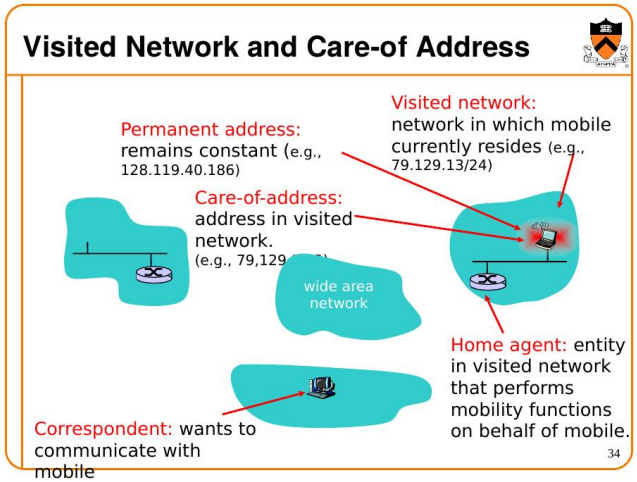
An IP address does two jobs at once: it says **who** you are and **where** you are (its network part decides where routers send packets). A phone that moves to another network therefore breaks routing: packets still go to the old network. If it simply takes a new address, every ongoing connection breaks (a connection is identified by IP addresses plus ports). **Mobile IP separates the two jobs:** a permanent address says who you are, a temporary address says where you are right now.

↳ *[D3] "Direct vs Indirect Mobile IP". The network/host split of an address is from the subnetting topic (Part B). A connection being identified by IPs and ports is from Part D.*

A.2 Vocabulary (slides 33 and 34)



Lecture slide 33



Lecture slide 34

Term	Meaning	Slide example
Home network	The mobile's permanent home	128.119.40/24
Permanent (home) address	Address in the home network; never changes; the correspondent always uses it	128.119.40.186
Home agent	Router on the home network that does mobility work while the mobile is away	(in home network)
Visited (foreign) network	Where the mobile currently is	79.129.13/24
Care-of address	Temporary address in the visited network	79.129.13.2
Foreign agent	Helper router in the visited network that receives tunnelled packets and delivers them	(in visited network)
Correspondent	The host that wants to talk to the mobile	

Mnemonic: permanent address = **who you are**. Care-of address = **where you can be found right now.** Mail analogy: you write to a friend who moves a lot. Option 1: he tells you each new address and you write to him directly (**direct routing**). Option 2: you write to his parents' fixed address and they forward it (**indirect routing**). Parents = home agent.

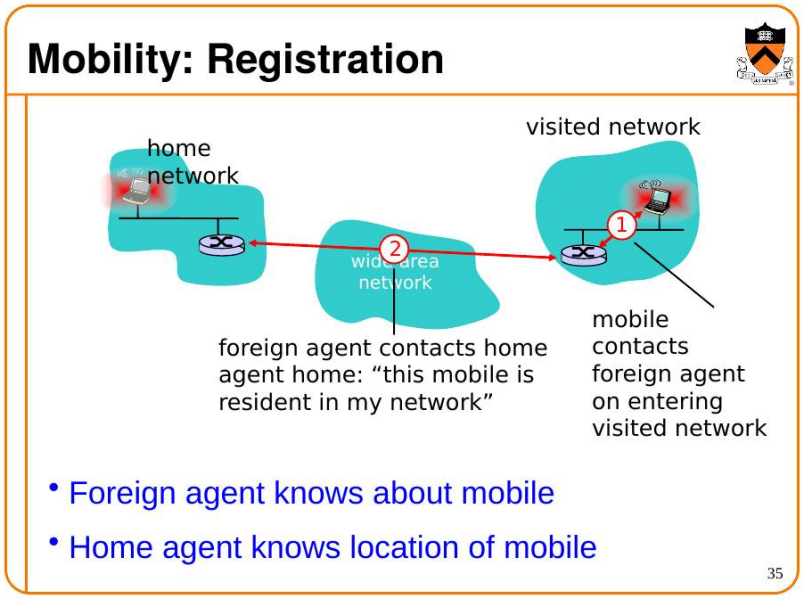
Slide typo: slide 34 calls the visited-network helper "home agent". It is the **foreign agent**. The home agent is always at home.

Collocated care-of address (slide 37, "mobile may perform the foreign agent functions"): the mobile gets a care-of address itself (for example by DHCP) and unwraps tunnelled packets itself. Advantage: works on any network, even one with no foreign agent. Cost: extra software on the mobile.

↳ *Getting an address by DHCP reuses the **DORA** process (Part B/C). Mobile IP's four goals: keep the home address anywhere; no change to fixed hosts or routers; avoid big detours; no overhead when at home.*

A.3 The three phases

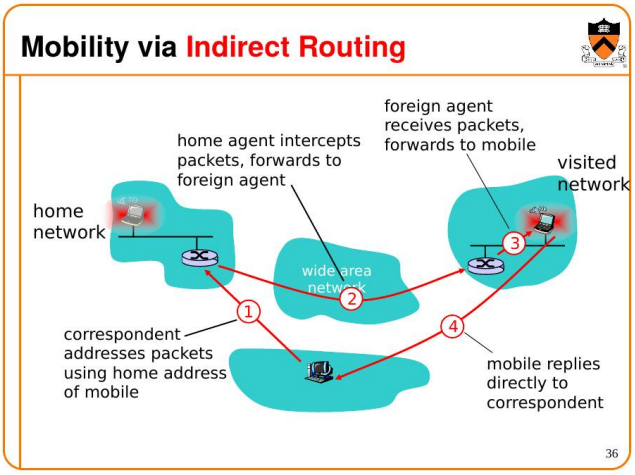
Phase	What happens	Detail
1. Agent discovery	Mobile learns its home agent before leaving, and finds a foreign agent (or gets its own care-of address) on arrival	Reuses **ICMP router advertisement/solicitation** messages; no new packet type
2. Registration (slide 35)	(1) Mobile contacts the foreign agent. (2) The foreign agent tells the home agent: "this mobile is resident in my network."	Request/Reply over **UDP port 434**; includes authentication; expires and must be renewed; cancelled on return home
3. Data transfer (slides 36 to 38)	Indirect or direct routing (below)	



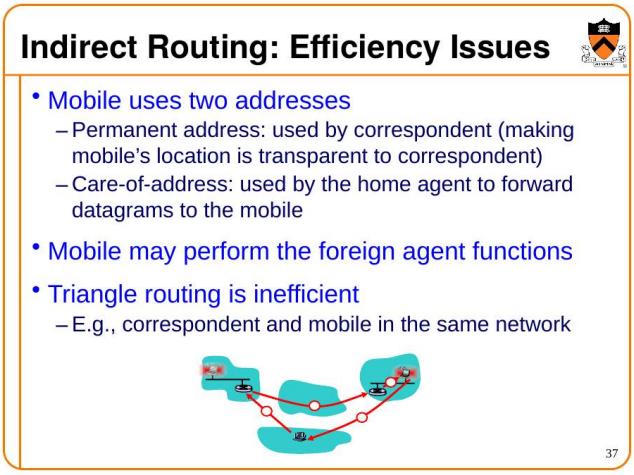
Lecture slide 35

*↳ UDP and a port number: Part D [T2] port mappings. Authentication stops someone **masquerading** as the mobile to hijack its traffic (A.8, active attacks).*

A.4 Indirect routing (slides 36 and 37)



Lecture slide 36



Lecture slide 37

1. The **correspondent** addresses packets to the mobile's **home address** (it does not know the mobile moved).
2. The **home agent intercepts** them (it answers for the mobile on the home LAN, "proxy ARP") and **tunnels** them to the foreign agent.
3. The **foreign agent** removes the tunnel wrapper and forwards to the mobile.
4. The **mobile replies directly** to the correspondent, using its home address as source.

Tunnelling = wrapping the whole original IP packet inside a new IP packet addressed to the foreign agent / care-of address; the far end removes the outer header.

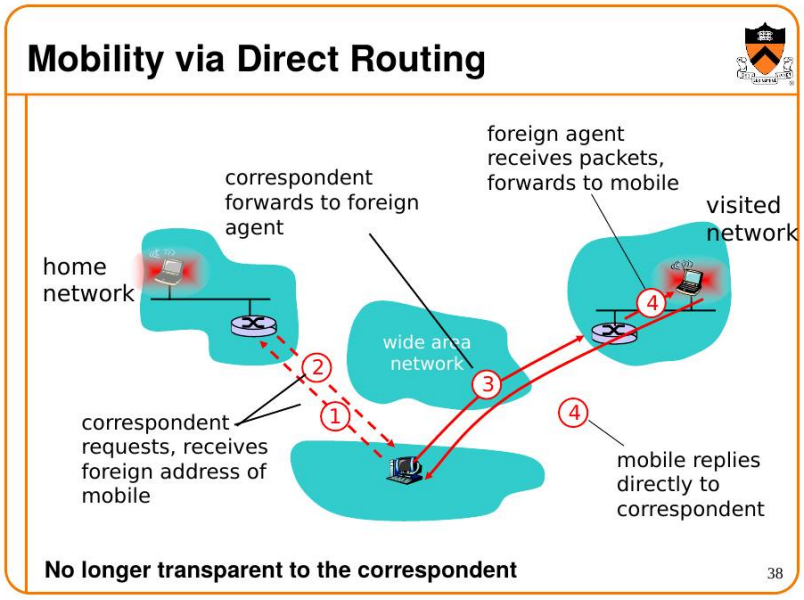
Step	From → To	Inner packet (src → dst)	Outer wrapper
1	Correspondent → home network	200.1.1.5 → 128.119.40.186	none
2	Home agent → foreign agent	200.1.1.5 → 128.119.40.186	home agent → foreign agent
3	Foreign agent → mobile (79.129.13.2)	200.1.1.5 → 128.119.40.186	removed
4	Mobile → correspondent	128.119.40.186 → 200.1.1.5	none

(The correspondent address 200.1.1.5 is illustrative; the others are from the slides.)

Efficiency (slide 37): the mobile has **two addresses**: the permanent one (used by the correspondent, which hides the mobile's location) and the care-of one (used by the home agent). **Triangle routing** is inefficient: forward packets go correspondent → home agent → mobile instead of straight across; replies go direct. Worst case ("double crossing"): correspondent and mobile are in the **same** visited network, yet packets travel to the home network and back.

↳ *Encapsulation (a packet inside a packet) is the same idea as tunnelling IPv6 over IPv4 in Part B. Proxy ARP: ARP is in Part B.* ***Triangle routing latency*** *is named in [D3] and Exam 2024 Q1.5.*

A.5 Direct routing (slide 38)



Lecture slide 38

1. The **correspondent asks for, and receives, the mobile's foreign (care-of) address** from the home agent.
2. It **sends straight to the foreign agent**.
3. The foreign agent forwards to the mobile.
4. The mobile replies directly.

Cost: **no longer transparent to the correspondent**; it must know about Mobile IP, and be updated if the mobile moves again.

	Indirect routing	Direct routing
Correspondent sends to	Mobile's home address	Foreign agent (after asking for the address)
Path	Correspondent → home agent → foreign agent → mobile (**triangle**)	Correspondent → foreign agent → mobile
Latency	Higher	**Lower**
Transparent to correspondent?	**Yes**	**No**
Correspondent must support Mobile IP?	No	**Yes**
Mail analogy	Letters go via the parents	Friend gives you his address

↳ *This table is the "latency comparison" mark in* ***Exam 2024 Q1.5*** *(7 marks) and [T2] 2024 Q1.4.*

A.6 Impact on higher layers (slide 39)

Wireless and mobility change path properties: **loss that is not congestion**, transient disruptions, changing RTT. The best-effort model is unchanged and TCP/UDP still run, but TCP suffers:

TCP behaviour	Why it hurts on wireless/mobile
Treats **every loss as congestion**	A wireless loss is usually a bit error; TCP slows down for nothing
Estimates **RTT** to set retransmission timers	A handoff or longer path (triangle routing) changes RTT suddenly, so timers are wrong
Poor with **out-of-order** packets	Different paths during handoff reorder packets; TCP may read it as loss

↳ *[D4] "packet reordering, congestion vs flow control". See Part D (wireless issues slide) for the strategy: the wireless link uses **ARQ** to mask errors.*

A.7 Security: the CIA triad (slide 41)

Principle	Meaning	Violated when
Confidentiality	Only authorised parties can read the data	Eavesdropping
Integrity	Data has **not been altered or forged**	Modification, replay
Availability	The service is usable when needed	Denial of service

Note: the speaker notes define integrity as honesty and moral principles; that is the everyday word. In the exam, integrity means **data not tampered with**.

↳ *[T2] 2026 Q3.3 asks "which CIA principle" (Availability).*

A.8 Passive vs active attacks (slides 42 to 44)

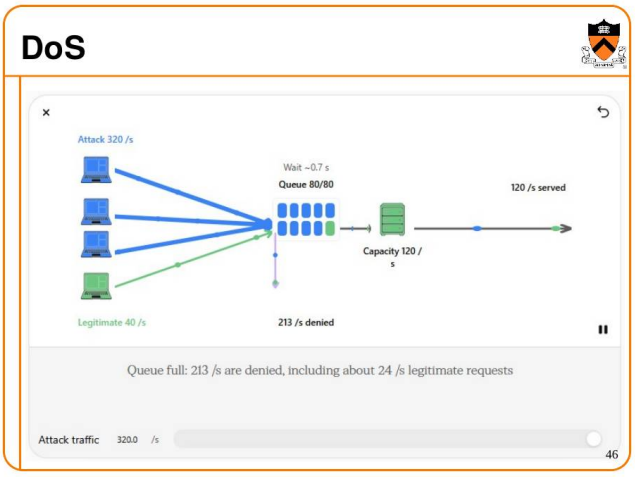
	Passive	Active
What	Eavesdrop; **does not alter** data	**Alter, disrupt or fake**
Kinds	Release of message contents; **traffic analysis** (frequency and length of messages, even if encrypted)	**Masquerade**, **replay**, **modification**, **denial of service**
Detect	**Difficult** (nothing changes)	**Easy** (visible effects)
Prevent	**Can be prevented** (encryption, secure protocols)	**Hard to prevent**
Threat to	Confidentiality	Integrity and availability
Attack	Plain English	Example
Masquerade	Pretend to be someone else	Stolen credentials; fake bank site
Replay	Capture a valid message and resend it	Resending a payment request = **duplicate payment**
Modification	Change, delay or reorder messages	Altering a transaction amount
Denial of service	Overload so users cannot get in	Flooding a server

Why encryption does not stop traffic analysis: it hides *what* is said, not *that* it is said, how big, or how often. **Wireless makes eavesdropping easy:** signals travel through the air for anyone nearby.

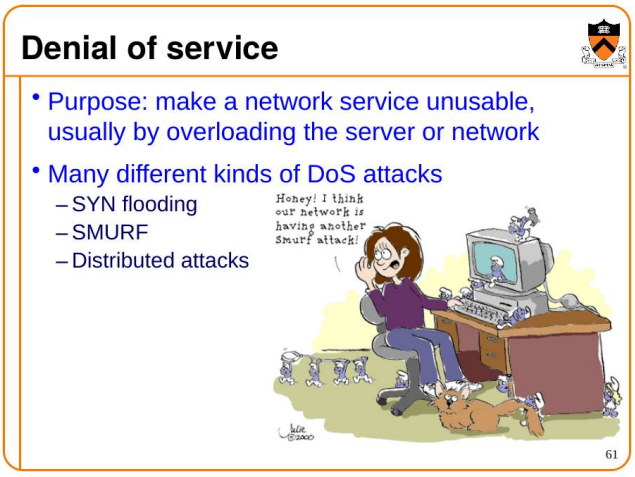
↳ *[D5] "Passive vs Active attacks", Test 2 2024 Q1.4. Broadcast nature of wireless: the Data link part (E).*

A.9 Denial of Service (slides 44 to 46 and 61 to 62)

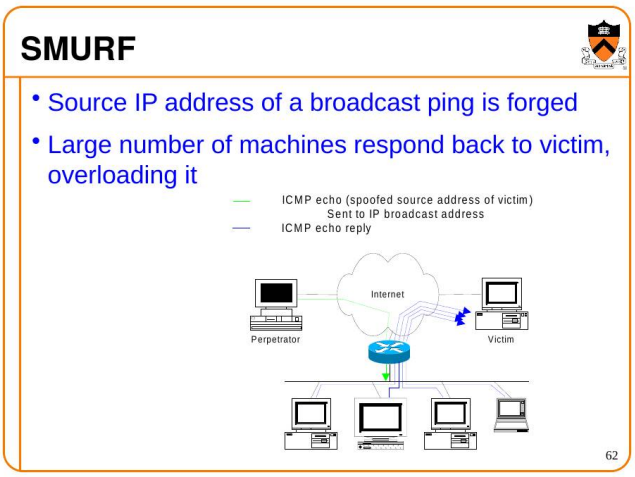
Definition (slide 61): make a service unusable, usually by overloading the server or network. Types named: **SYN flooding, SMURF, distributed attacks**.



Lecture slide 46: DoS



Lecture slide 61: Denial of service



Lecture slide 62: SMURF

SYN flooding. After sending SYN-ACK the server must remember its sequence number and keep a half-open connection. The attacker sends many SYNs and **never completes** the handshake, so the server fills up and refuses real clients. Defence: **SYN cookies** (the server derives its sequence number cryptographically from the client's address, port and a secret, then forgets it; if the handshake completes it can regenerate and verify the number).

SMURF (slide 62): the attacker pings a **broadcast address** with the **victim's address forged as the source**; every machine replies to the victim. One small packet becomes hundreds of replies (**amplification**).

DDoS: the flood comes from many machines at once (often compromised ones), so it is far harder to block.

Defences: ingress filtering (ISPs drop packets whose source address cannot be correct; stops forged-source attacks such as SMURF), **firewalls**, **rate limiting**, **IDS**.

↳ *SYN flood lives inside the **three-way handshake** you calculate in Part D. The broadcast address is the last address of a subnet (Part B). **Availability** is the CIA principle attacked.*

A.10 Symmetric encryption (slides 47 to 50)

Both parties share **one secret key**. The sender turns **plaintext** into **ciphertext** with the key; the receiver reverses it with the same key. Five ingredients: plaintext, encryption algorithm, secret key, ciphertext, decryption algorithm.

Security needs: (1) a **strong algorithm** (even with several ciphertext/plaintext pairs, the key cannot be worked out); (2) a **secure way to share the key**. Weakness: if the key leaks, everything is readable; this is the **key-distribution problem**.

Attacks: cryptanalysis (exploit the algorithm and plaintext patterns) and **brute force** (try every key). Slide example: a 4-letter password, 26 options per letter = 26^4 = **456,976** combinations; feasible if the site does not limit attempts. Defence: longer keys and attempt limits.

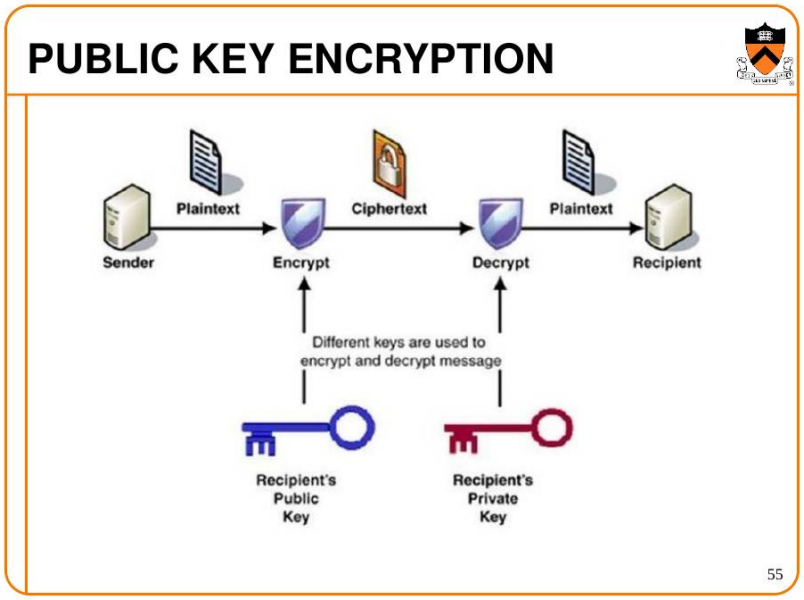
↳ *Counting possibilities is the same logic as host bits in subnetting (2^n minus 2), here 26^n.*

A.11 Public key (asymmetric) cryptography (slides 51 to 55)

A **key pair**: the **public key** is shared openly and **encrypts**; the **private key** is kept secret and **decrypts**. It underlies PKI: secure exchange, authentication and integrity checks.

1. Generate the key pair. 2. Exchange public keys. 3. The sender **encrypts with the recipient's public key**. 4. Send the ciphertext over any channel. 5. The recipient **decrypts with their own private key**.

> **Golden rule: encrypt with the RECIPIENT's PUBLIC key; decrypt with the RECIPIENT's PRIVATE key.**



Lecture slide 55: PUBLIC KEY ENCRYPTION

A, M, B diagram: A = sender, B = receiver, M = the open channel where an eavesdropper sits. Secrecy holds because ciphertext on M can only be opened with B's private key, which never leaves B.

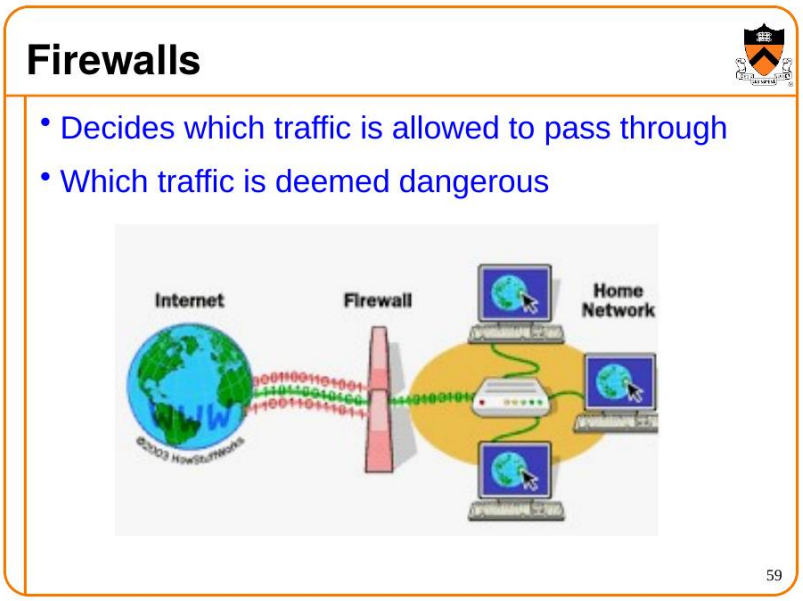
	Symmetric	Asymmetric
Keys	One shared secret	Public/private pair
Key distribution	**Hard**	**Easy** (publish the public key)
Speed	Fast	Slow (needs much longer keys)
Use	Bulk data	Key exchange, authentication

In practice, public-key crypto shares a key, then symmetric crypto carries the data. **HTTPS (443)** and **SSH (22)** use this combination.

↳ *[T2] 2026 Q3.2 (6 marks) and Exam 2023 Q2.5 (4 marks). Ports: Part D.*

A.12 Defences (slides 57 to 60)

Attack	Defence
Getting into the network	**Firewalls**
Software bugs, buffer overflows	**Intrusion Detection Systems**
Denial of service	**Ingress filtering, IDS**
TCP hijacking	**IPSec**
Packet sniffing	**Encryption (SSH, SSL, HTTPS)**
Social problems	**Education**



Lecture slide 59: Firewalls

Firewall: restricts traffic into, out of or within a private network; a gate that decides which traffic passes. **IDS:** monitors for malicious activity or policy violations. **Signature-based** compares packets with a database of known attack patterns (good for known attacks). **Anomaly-based** learns a baseline of normal behaviour and flags deviations (can catch new attacks).

Zero-day vulnerability (exam scope, not in this deck): a flaw unknown to the vendor, so **no patch exists**; an attacker who finds it first can exploit it before anyone can defend.

↳ *[D5] zero-day and proactive controls, Exam 2023 Q2.6 to 2.8. Firewalls also appear in the **Extranet** security challenge, [T1] Q1.4.4.*

A.13 Exam-ready answers for Part A

Direct vs indirect routing (Exam 2024 Q1.5, 7 marks). Define home network, home agent, foreign agent, care-of address, correspondent. Give the 4 steps of each method (A.4, A.5) with a diagram. Compare: indirect = triangle routing, higher latency, transparent; direct = shorter, lower latency, not transparent (correspondent must support Mobile IP). **Asymmetric encryption (Test 2 2026 Q3.2).** Draw A → [encrypt with B's public key] → ciphertext on M → [decrypt with B's private key] → B. Label A sender, M insecure channel, B receiver. Secrecy: only B's private key decrypts it and it is never shared. **DoS on a registrar database (Test 2 2026 Q3.3, 11 marks).** (i) DoS/DDoS, affects **Availability**. (ii) It floods the server so CPU, memory or connections run out and students cannot get through. (iii) Two controls: **firewall** filtering and **rate limiting** (or ingress filtering / IDS). **Handoff (Test 2 2026 Q3.1, 7 marks).** (i) **Handoff**: the base station sees the signal fade, asks neighbours how strongly they hear the phone, and transfers the connection to the strongest. (ii) Necessary because the user leaves the first cell; without it the download would drop. (iii) Two factors: signal strength (distance/obstacles), interference; speed of movement also counts.

PART B: The Network Layer

Source: Network Layer deck (81 slides) plus the subnetting exercises. Textbooks: Tanenbaum Ch. 5, Forouzan Ch. 18 to 22. **Exam tags:** [T2], [D3].

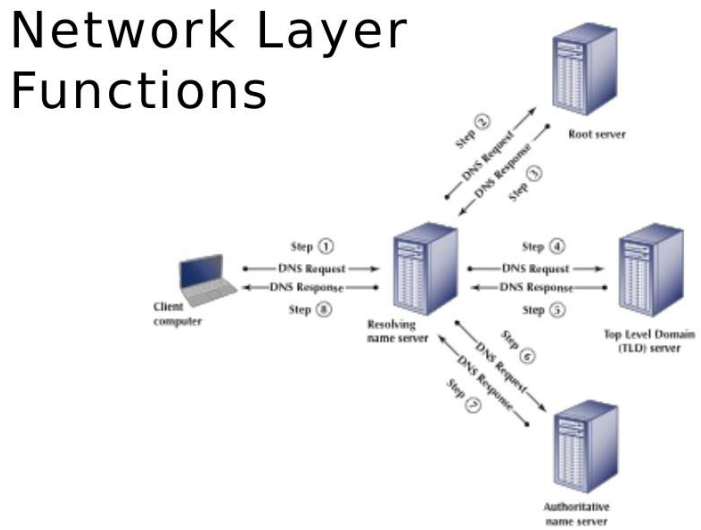
B.1 What the network layer does (slides 2 to 4)

It delivers packets **between endpoints over multiple links**, across several networks. The main protocol is **IP (Internet Protocol)**: it gives devices **logical (IP) addresses**, delivers packets from one network to another, and helps routers decide where packets go.

	IPv4	IPv6
Address size	32 bits	128 bits
Unique addresses	about 4.29 billion	about 3.4×10^{38}
Example	192.168.10.25	2001:0db8:85a3:0000:0000:8a2e:0370:7334

The slide's analogy: a town with only 100 houses; house 101 cannot exist. Fix: build another town (NAT, B.10) or a new addressing system (IPv6).

B.2 Addressing (slides 5 to 9)



Lecture slide 10: Network Layer Functions

Layer	Address type	Example	Size
Application	URL	www.indiana.edu	text
Network	IP address	129.79.78.193	4 bytes
Data link	MAC address	1C-6F-65-F8-33-8A	6 bytes

Addresses are **translated between layers**: **DNS** (name → IP) and **ARP** (IP → MAC of the next device).

Dotted-decimal notation: the 32 bits are split into four bytes, each written in decimal. Example: 10000001 01001111 01001110 11000001 = **129.79.78.193**. One byte = 8 bits, values 0 to 255.

Network part vs host part. Part of every IP address identifies the **network**, the rest the **host**. Two ways to say where the split is:

Method	How	Status		
Classful	First bits of the address fix the split	Discontinued, but names are still used		
CIDR (classless)	A **prefix length** or **subnet mask** says how many bits are network bits	Used today		
Class	Leading bits	First octet range	Default mask	Use

Method	How	Status		
A	0	1 to 126 (127 = loopback)	255.0.0.0 (/8)	Very large networks
B	10	128 to 191	255.255.0.0 (/16)	Medium networks
C	110	192 to 223	255.255.255.0 (/24)	Small networks
D	1110	224 to 239	n/a	Multicast
E	1111	240 to 255	n/a	Reserved

↳ *[D3] "Binary IPv4 class derivations" and 2019 Q3.1: look at the **leading high-order bits**. Classful waste (slide 68) is why CIDR (/19, /22...) replaced it (B.10).*

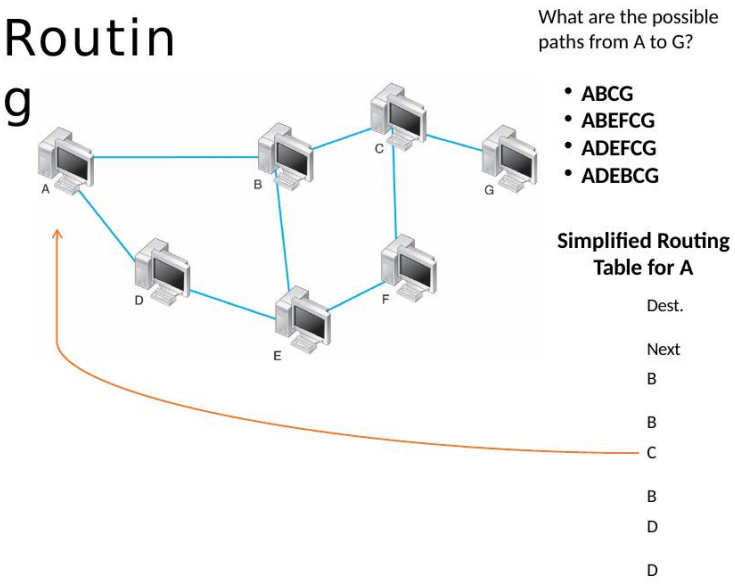
B.3 Dynamic addressing: DHCP DORA (slide 8, and 75)

Configuring each device by hand is slow, and permanent addresses are wasteful when devices are offline. A **DHCP server** hands out addresses automatically, **leased for a length of time**.

Step	Message	What happens
Discover	Client **broadcasts**	"Is there a DHCP server?"
Offer	Server → client	Offers an IP address and settings
Request	Client → server	"I'll take that address"
Acknowledge	Server → client	Confirms the lease

↳ *[D5] "DHCP DORA process". Slide 27 of the Transport deck (Part D.8) warns about **synchronization**: after a power cut all hosts ask the DHCP server at once. Mobile IP's collocated care-of address (Part A) is obtained this way.*

B.4 Routing: the basics (slides 11 to 15)



Lecture slide 12: Routing

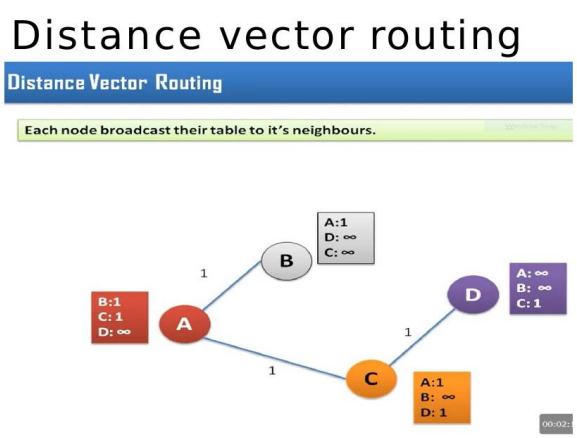
Routing is choosing the path a packet takes from source to destination. Routers keep **routing tables** (destination → next hop). Slide 12's example: from A to G the possible paths are ABCG, ABEFCG, ADEFCG, ADEBCG; the table for A says "to reach G, send to B".

Choice	Option 1	Option 2
Who decides	**Centralized**: one computer decides (not common now)	**Decentralized**: each node decides independently; routers exchange information (used by the Internet)
How tables are made	**Static**: fixed, set by hand	**Dynamic**: updated periodically by protocols

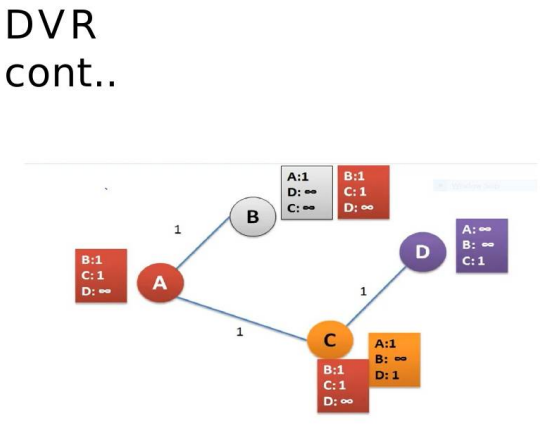
Dynamic algorithm families: Distance vector: based on the number of **hops**. **Link state**: based on hops, **circuit speed and traffic congestion**; gives more reliable, up-to-date paths.

B.5 Distance-vector routing, worked (slides 16 to 25)

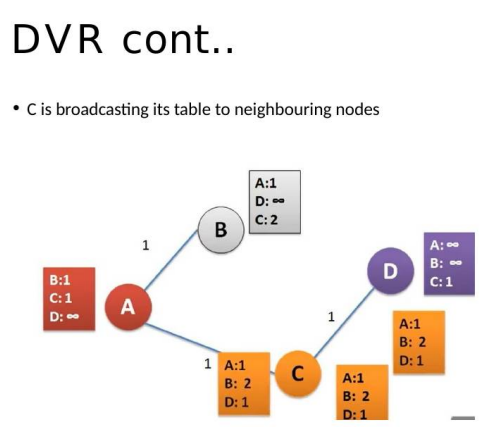
Idea in one line: every node tells its **neighbours** its whole table; each node updates its own table from what it hears (shortest distance wins).



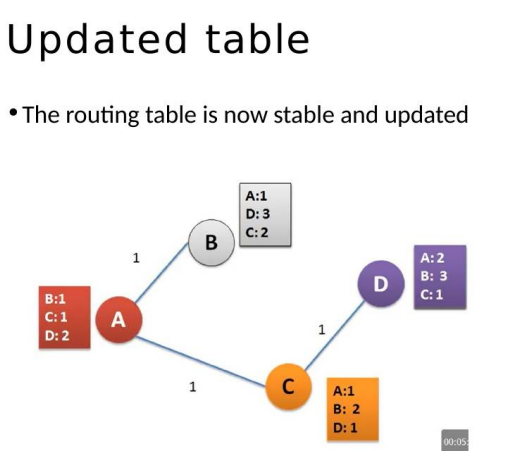
Lecture slide 16: Distance vector routing



Lecture slide 17: DVR cont..



Lecture slide 19: DVR cont..



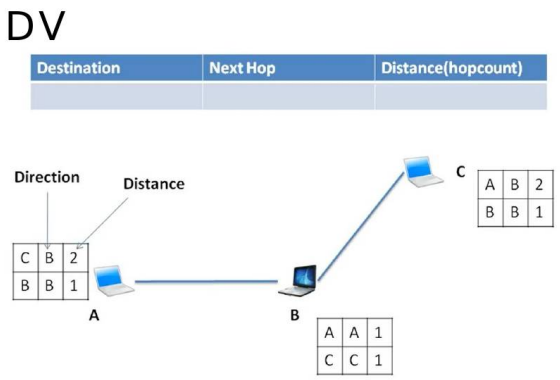
Lecture slide 21: Updated table

Walk through the slides' example (links A-B, A-C, C-D, each cost 1):

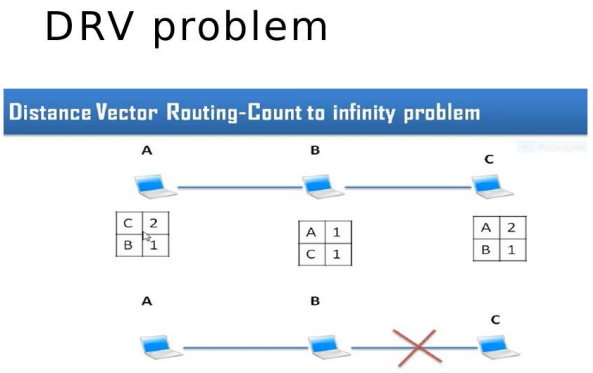
1. **Start:** each node knows only its direct neighbours (A: B=1, C=1, D=∞; and so on).
2. **Exchange:** nodes broadcast their tables to their neighbours (slide 19: C broadcasts its table). A learns D is reachable via C: A to D = 1 + 1 = **2**. B learns C is reachable via A: B to C = 1 + 1 = **2**.
3. **Repeat until nothing changes.** Final stable tables: A (B:1, C:1, D:2); B (A:1, C:2, D:3); C (A:1, B:2, D:1); D (A:2, B:3, C:1).

The rule at each step: **new distance = distance to the neighbour + the distance that neighbour advertises.**

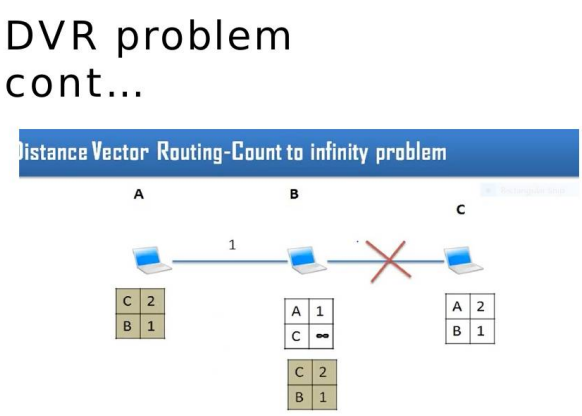
The problem: count to infinity (slides 22 to 25). Chain A - B - C. B reaches C directly (cost 1); A reaches C via B (cost 2). Now the **B-C link fails**. B sets C = ∞, but before it advertises that, A advertises "C = 2" to B. B believes it can reach C via A (cost 2 + 1 = 3) while A still thinks it goes via B: a **routing loop**; the cost keeps increasing step by step ("counting to infinity") and packets loop.



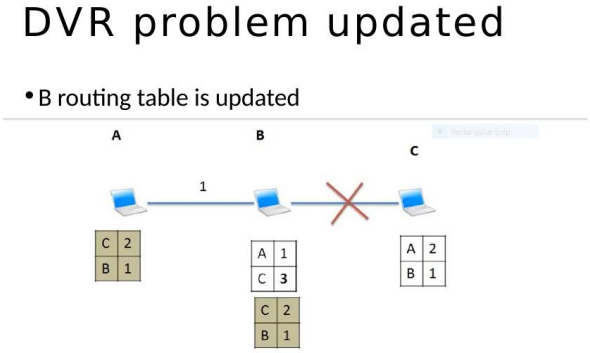
Lecture slide 22: DVR



Lecture slide 23: DRV problem



Lecture slide 24: DVR problem cont...



Lecture slide 25: DVR problem updated

↳ Distance vector is the family RIP belongs to (B.6). OSPF's link-state approach (each node learns the whole map) avoids this problem. The same "slow news" problem is why wireless/mobile routing uses on-demand protocols (MANET/DSR, [D3]).

B.6 Routing protocols (slides 26 to 29)

Protocol	Type	Used for	Key points
RIP (Routing Information Protocol)	Dynamic **distance vector**	**Interior** routing	Tables broadcast periodically (about every minute); counts hops, picks the shortest; good for **small, simple** networks
OSPF (Open Shortest Path First)	Dynamic **link state**	**Interior** routing	Most widely used on large enterprise networks; more reliable paths; less burden on the network because **only updates are sent**
EIGRP (Cisco)	Dynamic link state (as the slide says)	Interior	Records capacity, delay, reliability and load for all paths; also uses its neighbours' tables
BGP (Border Gateway Protocol)	Dynamic DV	**Exterior** routing between networks	Far more complex; gives routing information only on selected (preferred/best) routes; lets networks using different internal protocols communicate

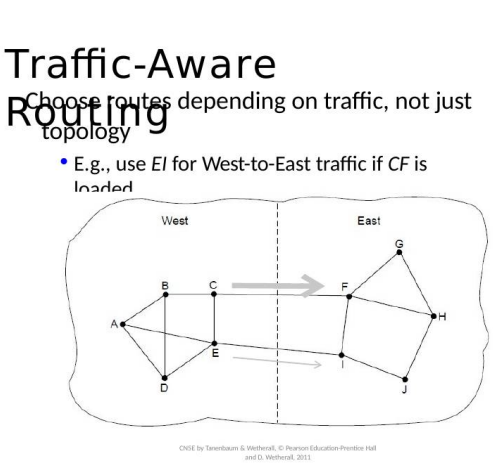
*↳ "Interior" = inside one organisation; "exterior" = between organisations. This is the question on slide 29: how can networks with different internal protocols talk? **BGP.***

B.7 Congestion control (slides 31 to 38)

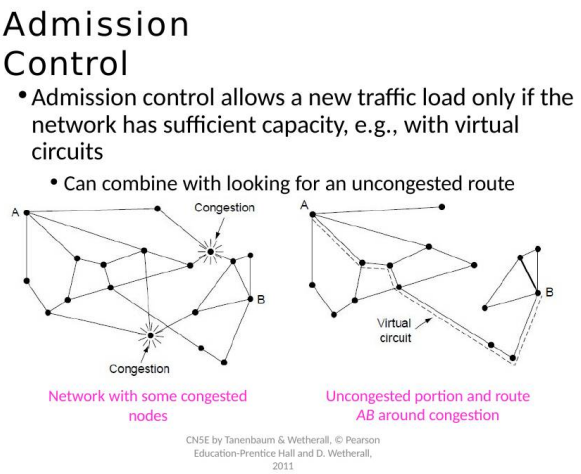
Congestion = too much traffic offered, so performance drops (loss and retransmissions). It also arises at low-bandwidth links or slow routers. **Flow control** = sender ↔ receiver pace; **congestion control** = making sure the **network** can carry the offered load. Network and Transport layers work together; the network layer does its best with the load, nodes should also **reduce** the offered load (Transport).

Approach (slides)	Idea
Traffic-aware routing (34)	Choose routes by current traffic, not only topology; avoid oscillation

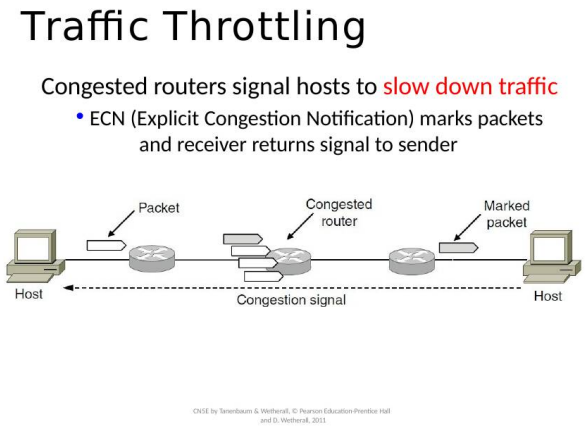
Approach (slides)	Idea
Admission control (35)	Accept a new flow only if the network has capacity (for example virtual circuits); route around congested nodes
Traffic throttling (36)	Congested routers signal hosts to slow down; **ECN** marks packets and the receiver returns the signal
Load shedding (37, 38)	When all else fails, **drop packets**. Link-by-link gives rapid relief; end-to-end takes longer but targets the cause



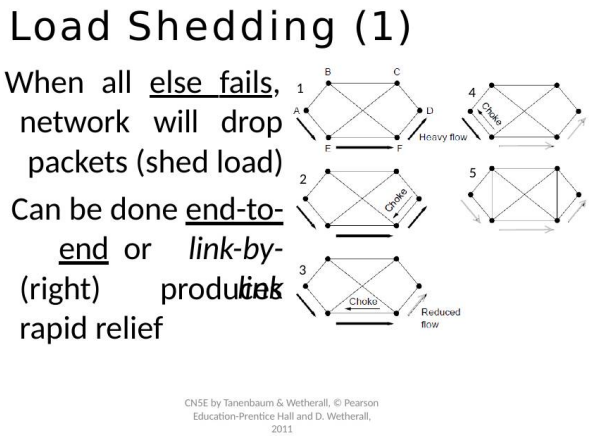
Lecture slide 34: Traffic-Aware Routing



Lecture slide 35: Admission Control



Lecture slide 36: Traffic Throttling



Lecture slide 37: Load Shedding (1)

↳ *Transport-layer half of congestion control: Part D. [D4] "congestion vs flow control".*

B.8 Quality of Service (slides 41 to 50)

QoS = the network manages traffic so each application gets the performance it needs. If a WhatsApp video call, a 5 GB download and an email compete, **which gets priority?** Four issues: what applications need; how to regulate traffic entering; how to reserve router resources; whether the network can safely accept more.

Technique	Idea
Application requirements (44)	Different apps care about different things (delay, jitter, bandwidth, loss)
Traffic shaping (45)	Regulate the **average rate and burstiness** of data entering the network; allows guarantees
Packet scheduling (46)	Divide link resources among flows with alternatives to FIFO (for example **round-robin**)
Admission control (47)	Take a **flow specification** and decide if the network can carry it
Integrated Services (48, 49)	QoS **per flow**, handles multicast; **RSVP**: receiver requests back toward the sender, each router reserves resources, requests for the same flow merge; whole path or no reservation
Differentiated Services (50)	**Classes** of QoS; the expedited class is sent in preference to the regular class

Application Requirements (1)

Different applications care about different properties

- We want all applications to get what they need

Application	Bandwidth	Delay	Jitter	Loss
Email	Low	Low	Low	Medium
File sharing	High	Low	Low	Medium
Web access	Medium	Medium	Low	Medium
Remote login	Low	Medium	Medium	Medium
Audio on demand	Low	Low	High	Low
Video on demand	High	Low	High	Low
Telephony	Low	High	High	Low
Videoconferencing	High	High	High	Low

"High" means a demanding requirement, e.g., low delay

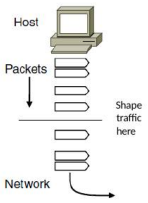
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Lecture slide 44: Application Requirements (1)

Traffic Shaping (1)

Traffic shaping regulates the average rate and burstiness of data entering the network

- Lets us make guarantees

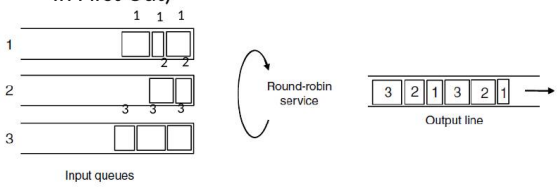


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Lecture slide 45: Traffic Shaping (1)

Packet Scheduling (1)

Packet scheduling divides router/link resources among traffic flows with alternatives to FIFO (First In First Out)

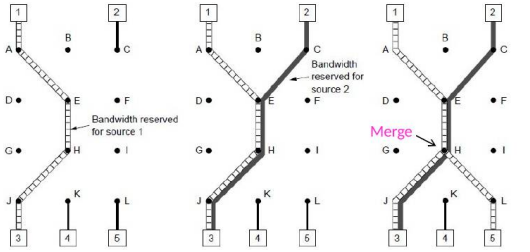


Example of round-robin queuing

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Lecture slide 46: Packet Scheduling (1)

Integrated Services (2)



R3 reserves flow from S1

R3 reserves flow from S2

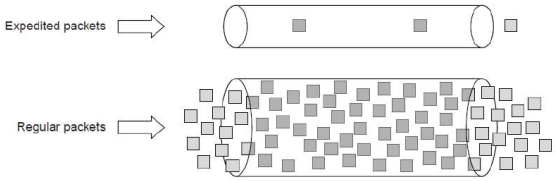
R5 reserves flow from S1; merged with R3 at H

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Lecture slide 49: Integrated Services (2)

Differentiated Services (1)

- Design with classes of QoS; customers buy what they want
  - Expedited class is sent in preference to regular class
  - Less expedited traffic but better quality for applications



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Lecture slide 50: Differentiated Services (1)

B.9 Internetworking (slides 51 to 59)

Internetworking = connecting different networks into one larger network, even if they differ in technology, addressing, packet sizes, protocols and QoS. Example: a student in residence reaches a server on the library network. **IP carried all the way** is what lets the packet continue across different underlying technologies.

How Networks Differ

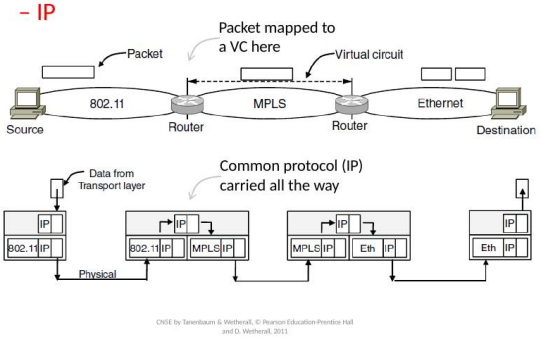
- Various networks use differ technologies, addressing methods, packet sizes, protocols, and QoS mechanisms.

Item	Some Possibilities
Service offered	Connectionless versus connection oriented
Addressing	Different sizes, flat or hierarchical
Broadcasting	Present or absent (also multicast)
Packet size	Every network has its own maximum
Ordering	Ordered and unordered delivery
Quality of service	Present or absent; many different kinds
Reliability	Different levels of loss
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, packet, byte, or not at all

Lecture slide 52: How Networks Differ

How Networks Can Be Connected

Internetworking based on a common network layer – IP

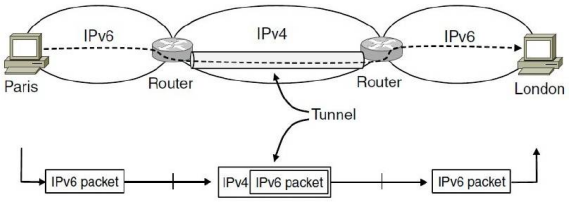


Lecture slide 54: How Networks Can Be Connected

Tunneling (1)

Connects two similar networks through a middle one

- Packets are encapsulated over the middle
- Imagine two company IPv6 networks need to communicate, but the network between them supports only IPv4. The IPv6 packet can be placed inside an IPv4 packet to cross the IPv4 network. At the other end, the IPv4 header is removed and the original IPv6 packet continues normally.

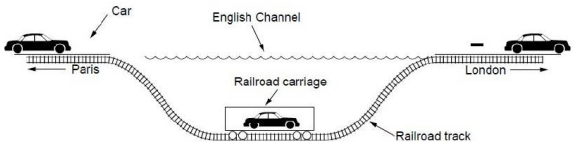


Lecture slide 55: Tunneling (1)

Tunneling (2)

Tunneling analogy:

- tunnel is a link; packet can only enter/exit at ends



Lecture slide 56: Tunneling (2)

Tunnelling (55, 56): connects two similar networks through a different one in the middle by **encapsulating** packets. Example: two IPv6 company networks separated by an IPv4-only network: the IPv6 packet is put inside an IPv4 packet, and the IPv4 header is removed at the far end. The tunnel acts like a link: packets enter and exit only at its ends. (Mobile IP uses this idea, Part A.)

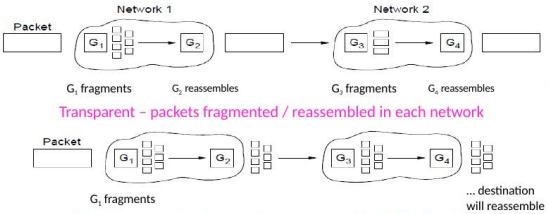
Fragmentation (57 to 59): a large packet that must cross a network with a smaller maximum size is cut into **fragments**.

Type	What happens
Transparent	Packets fragmented and reassembled inside each network
Non-transparent	Fragments stay separate and are **reassembled at the destination** (IP style)

IP-style fragments carry a **packet number, start offset and an end bit** so the destination can put them back. **Path MTU discovery** avoids fragmentation: find the largest size that fits the whole path; a router that cannot forward a packet drops it and tells the sender the MTU; the sender shrinks and retries (the slide's "try 1200 / try 900").

Packet Fragmentation (1)

- Occurs when a large packet must travel through a network that supports a smaller max packet size
- A large packet is divided into smaller pieces called fragments.
- Transparent fragmentation- packets maybe reassembled before entering next network
- Non-transparent, fragments remain separate and reassembled in the destination



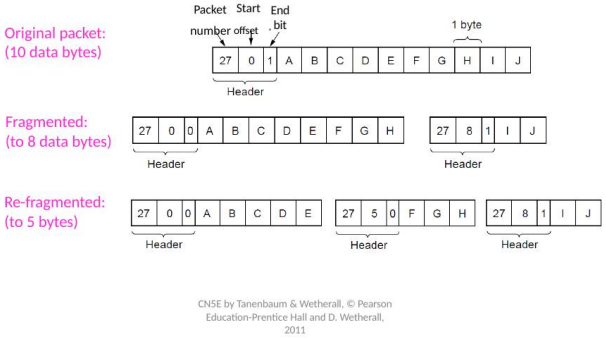
Transparent – packets fragmented / reassembled in each network

Non-transparent – fragments are reassembled at destination

Lecture slide 57: Packet Fragmentation (1)

Packet Fragmentation (2)

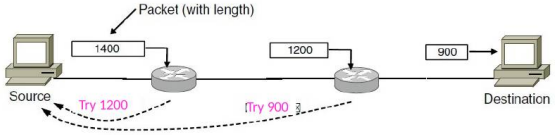
- Example of IP-style fragmentation:



Lecture slide 58: Packet Fragmentation (2)

Packet Fragmentation (3)

- Path MTU discovery: a technique used to avoid fragmentation
- Determines the largest packet size that can travel across the entire path from S to D
- If a packet is too large for one of the networks along the route
- The router discards it & informs S about Max Trans Unit
- Sender will then reduce the packet size & tries again



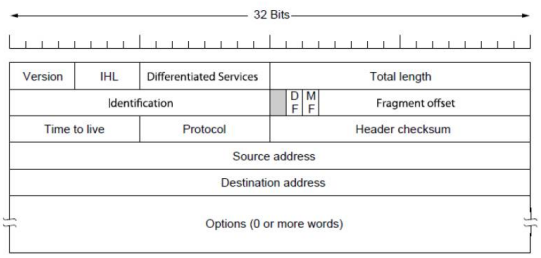
Lecture slide 59: Packet Fragmentation (3)

B.10 IP in the Internet (slides 60 to 73)

Design principles of IP: make sure it works, keep it simple, exploit modularity, expect heterogeneity (a smartphone on Wi-Fi, a university on Ethernet and an ISP on fibre are all joined by IP).

IP Version 4 Protocol (1)

- IPv4 header attached to every packet
- Has infor routers need to deliver & manage packet
- Routers examine relevant fields in this header when forwarding the packet toward its destination.

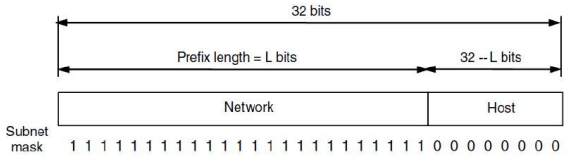


Lecture slide 63: IP Version 4 Protocol (1)

IP Addresses (1) – Prefixes

Addresses are allocated in blocks called prefixes

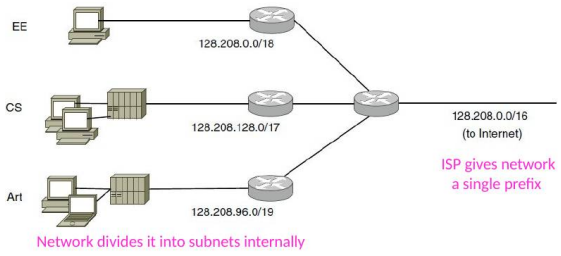
- A prefix is determined by the network portion
- Remaining bits are used as host addresses
- Prefix is written using the format 192.168.1.0/24
- Host bits being 8, 2^8=256 possible addresses
- Block ranges from 192.168.1.0 – 192.168.1.255



Lecture slide 64: IP Addresses (1) – Prefixes

IP Addresses (2) – Subnets

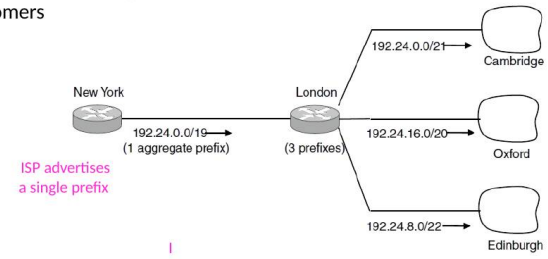
- Dividing a large IP address block into smaller networks called subnets.
- An organization receives one IP prefix from the ISP
- Internally can be divide the prefix among the various departments



Lecture slide 65: IP Addresses (2) – Subnets

IP Addresses (3) – Aggregation

- Process of combining several small IP prefixes into one large prefix
- To reduce the size of routing tables
- Instead of ISP advertising separate routes for every customer network,
- It advertises one larger prefix that represents those customers



Lecture slide 66: IP Addresses (3) – Aggregation

IPv4 header: attached to every packet; carries what routers need to deliver and manage it (slide 63).

Prefixes (slide 64): addresses are allocated in blocks called prefixes, written like **192.168.1.0/24**. The network part is the prefix; the remaining bits are host bits. 8 host bits gives 2^8 = 256 addresses (192.168.1.0 to 192.168.1.255).

Subnets (65): divide a large block into smaller networks, for example one prefix from the ISP shared among departments.

Aggregation (66): combine several small prefixes into **one larger prefix** to shrink routing tables; the ISP advertises one prefix for all its customers.

Longest matching prefix (67): when several entries match a destination, the router picks the **longest (most specific)** one. Slide example: route A = 192.24.0.0/16 and route B = 192.24.32.0/24; a packet for **192.24.32.50** matches both, but /24 is more specific, so it goes to **router B**.

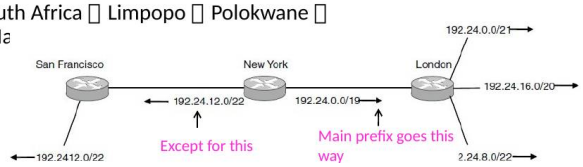
IP Addresses (4) – Longest Matching Prefix

- How router decides which routing-table entry to use when more prefixes matches the D IP address.
- Router selects longest matching prefix

Suppose router has two routes router A- 192.24.0.0/16 & router B – 192.24.32.0/24

A packet arrives for 192.24.32.50 both routes match the D, but /24 is more specific than 16 thus the router sends the packet to router B.

South Africa Limpopo Polokwane Ma

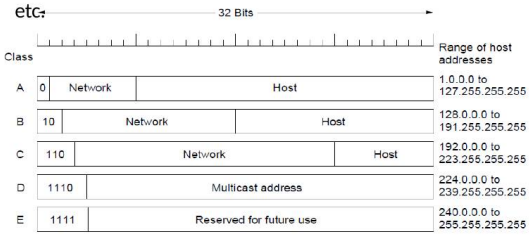


Lecture slide 67: IP Addresses (4) – Longest Matching Prefix

IP Addresses (5) – Classful Addresing

Old addresses came in blocks of fixed size (A, B, C)

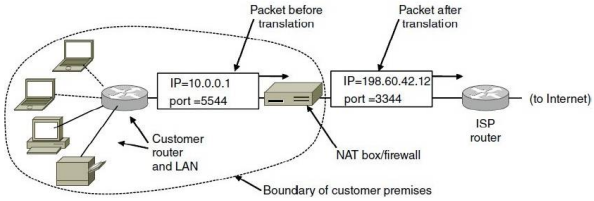
- Carries size as part of address, but lacks flexibility
- Called classful (vs. classless) addressing
- Issue is that these fixes are inflexible & could waste large number of IP addresses
- Which is why classless addressing was preferred using /19, /22 etc.



Lecture slide 68: IP Addresses (5) – Classful Addresing

IP Addresses (6) – NAT

- Allows devices on a private network to share one public IP address when comm. with the internet
- NAT device (router) translates devices' private addresses & and the public IP address.
- Several nodes share same public address, NAT also uses TCP/UDP port numbers to vary their connections



Lecture slide 69: IP Addresses (6) – NAT

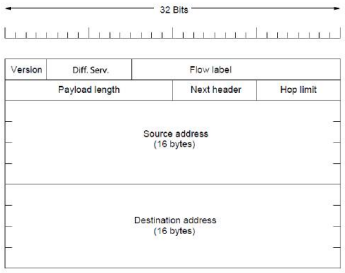
Classful vs classless (68): old fixed-size blocks (A, B, C) waste addresses; classless addressing (/19, /22 and so on) is flexible.

NAT (69, 70): devices on a private network **share one public IP**. The NAT router translates private addresses and uses **TCP/UDP port numbers** to tell connections apart. Slide example: laptop 192.168.1.10 and phone 192.168.1.20 go out through one public address, mapped to ports 1111 and 2222; replies are matched in the mapping table and sent to the right device.

IPv6 (71 to 73): designed to fix address exhaustion and also support billions of hosts, smaller routing tables, simpler protocol, better security, type of service, multicasting, roaming without changing address, and coexistence with IPv4. Its header has longer addresses (128 bits) but is **simpler** because other functions move to **extension headers**.

IP Version 6 (2)

IPv6 protocol header has much longer addresses (128 vs. 32 bits) and is simpler (by using extension headers)



Lecture slide 72: IP Version 6 (2)

IP Version 6 (3)

IPv6 extension headers handles other functionality

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

Lecture slide 73: IP Version 6 (3)

↳ *Mobile IPv6 (roaming without changing address) is the "roaming host" goal here. NAT's use of ports links to Part D.*

B.11 Internet control protocols and MPLS (slides 75 to 79)

Protocol	Job
ICMP	Companion to IP that **returns error information**; also used by **traceroute**
ARP	Finds the **Ethernet (MAC) address of a local IP address**: host asks, owner replies
DHCP	Assigns a local IP to a host; the host sends a request, the server grants a **lease**
MPLS	**Label switching**: packets follow established paths indicated by a **label** added below the IP layer when entering an MPLS network (for example an ISP) and removed on leaving; used for QoS

Internet Control Protocols (2)

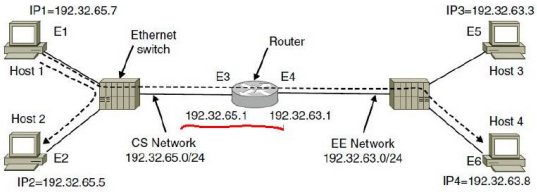
Main ICMP (Internet Control Message Protocol) types:

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo and Echo reply	Check if a machine is alive
Timestamp request/reply	Same as Echo, but with timestamp
Router advertisement/solicitation	Find a nearby router

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Lecture slide 76: Internet Control Protocols (2)

Internet Control Protocols (3)



Frame	Source IP	Source Eth.	Destination IP	Destination Eth.
Host 1 to 2, on CS net	IP1	E1	IP2	E2
Host 1 to 4, on CS net	IP1	E1	IP4	E3
Host 1 to 4, on EE net	IP1	E4	IP4	E6

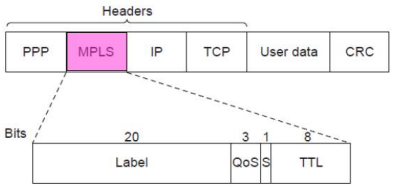
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Lecture slide 77: Internet Control Protocols (3)

Label Switching and MPLS (1)

MPLS (Multi-Protocol Label Switching) sends packets along established paths; ISPs can use for QoS

- Path indicated with label below the IP layer

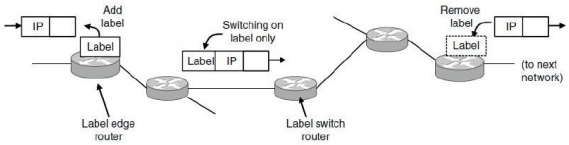


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Lecture slide 78: Label Switching and MPLS (1)

Label Switching and MPLS (2)

Label added based on IP address on entering an MPLS network (e.g., ISP) and removed when leaving it



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Lecture slide 79: Label Switching and MPLS (2)

↳ ARP is the "MAC address resolution" step that completes [T1] Layer 2 delivery. ICMP router advertisements are what Mobile IP reuses (Part A).

B.12 Subnetting: the method you need for [T2]

Everything depends on powers of two. With **h host bits**: usable hosts = $\mathbf{2^h - 2}$ (minus the network address and the broadcast address). Prefix = **32 − h**.

Host bits h	2	3	4	5	6	7	8	9	10
Usable hosts	2	6	14	30	62	126	254	510	1022
Prefix	/30	/29	/28	/27	/26	/25	/24	/23	/22
Block size (last octet)	4	8	16	32	64	128	256	n/a	n/a
Mask (last octet)	252	248	240	224	192	128	0	n/a	n/a

FLSM (fixed-length): every subnet the same size. Subnets = $2^{\text{borrowed bits}}$. **VLSM (variable-length):** each subnet sized to its need. **Manual VLSM, step by step:**

1. **Sort** the requirements **largest first** (descending).
2. For each, find the smallest h with $2^h - 2 \geq$ hosts; prefix = 32 − h.
3. Start at the parent's network address. **Network** = start. **Broadcast** = start + block size − 1. **First usable** = network + 1. **Last usable** = broadcast − 1.
4. **Next subnet starts at the previous broadcast + 1.** Repeat.
5. The next address after the last allocation is the **first unallocated** address.

Junior-technician traps: assigning the network or broadcast address to a host; putting a host in the wrong range; **FLSM /27 for unequal subnets wastes addresses**.

Worked example 1: FLSM, 172.20.0.0 with mask /27 (Transport deck exercise 1)

Class B, 172.20.0.0 /27: subnet bits = 27 − 16 = **11**, so 2^{11} = **2048 subnets**; host bits = 5, so $2^5 - 2$ = **30 usable hosts** each. First five subnets (the 5 departments):

Subnet #	Subnet ID	Mask	First usable	Last usable	Broadcast
0	172.20.0.0/27	255.255.255.224	172.20.0.1	172.20.0.30	172.20.0.31
1	172.20.0.32/27	255.255.255.224	172.20.0.33	172.20.0.62	172.20.0.63
2	172.20.0.64/27	255.255.255.224	172.20.0.65	172.20.0.94	172.20.0.95
3	172.20.0.96/27	255.255.255.224	172.20.0.97	172.20.0.126	172.20.0.127
4	172.20.0.128/27	255.255.255.224	172.20.0.129	172.20.0.158	172.20.0.159

Worked example 2: VLSM, 172.20.0.0/24 (Transport deck example 2)

Computer science labs 60, lecture hall 30, admin office 12, server room 4:

Subnet	Hosts needed	Prefix / mask	Network address	First usable	Last usable	Broadcast
CS labs	60	/26 255.255.255.192	172.20.0.0	172.20.0.1	172.20.0.62	172.20.0.63
Lecture hall	30	/27 255.255.255.224	172.20.0.64	172.20.0.65	172.20.0.94	172.20.0.95
Admin	12	/28 255.255.255.240	172.20.0.96	172.20.0.97	172.20.0.110	172.20.0.111
Server room	4	/29 255.255.255.248	172.20.0.112	172.20.0.113	172.20.0.118	172.20.0.119

First unallocated address: 172.20.0.120 (addresses left in 172.20.0.0/24: 136)

Worked example 3: Test 2 2026 Q1 (15 marks), 10.10.0.0/16

LAN1 300, LAN3 100, LAN4 55, LAN2 30, WAN1 2. **1.1:** host bits in a /16 = 32 − 16 = **16**, so 2^{16} = 65,536 addresses.

Subnet	Hosts needed	Prefix / mask	Network address	First usable	Last usable	Broadcast
LAN1	300	/23 255.255.254.0	10.10.0.0	10.10.0.1	10.10.1.254	10.10.1.255
LAN3	100	/25 255.255.255.128	10.10.2.0	10.10.2.1	10.10.2.126	10.10.2.127
LAN4	55	/26 255.255.255.192	10.10.2.128	10.10.2.129	10.10.2.190	10.10.2.191
LAN2	30	/27 255.255.255.224	10.10.2.192	10.10.2.193	10.10.2.222	10.10.2.223

Subnet	Hosts needed	Prefix / mask	Network address	First usable	Last usable	Broadcast
WAN1	2	/30 255.255.255.252	10.10.2.224	10.10.2.225	10.10.2.226	10.10.2.227

First unallocated address: 10.10.2.228 (addresses left in 10.10.0.0/16: 64796)

> **1.5 Technician trap:** he assigns **10.10.2.150** to a device in LAN3. LAN3 is 10.10.2.0/25 (10.10.2.0 to 10.10.2.127). 150 is **outside** that range; it falls inside **LAN4 (10.10.2.128/26, up to .191)**. So the address does **not** belong to LAN3.

Worked example 4: Special Test 2 (August 2026) Q1, 172.20.0.0/23

Administration 120, Finance 60, Research 28, Security 12. (a) Administration needs $2^h - 2 \geq 120$, so h = 7, prefix **/25**.

Subnet	Hosts needed	Prefix / mask	Network address	First usable	Last usable	Broadcast
Administration	120	/25 255.255.255.128	172.20.0.0	172.20.0.1	172.20.0.126	172.20.0.127
Finance	60	/26 255.255.255.192	172.20.0.128	172.20.0.129	172.20.0.190	172.20.0.191
Research	28	/27 255.255.255.224	172.20.0.192	172.20.0.193	172.20.0.222	172.20.0.223
Security	12	/28 255.255.255.240	172.20.0.224	172.20.0.225	172.20.0.238	172.20.0.239

First unallocated address: 172.20.0.240 (addresses left in 172.20.0.0/23: 272)

Worked example 5: 2023 Q2.1, 192.168.0.0/24

HR 30, Sales 25, IT 15, Future growth 50 (allocate Future growth first, it is largest):

Subnet	Hosts needed	Prefix / mask	Network address	First usable	Last usable	Broadcast
Future growth	50	/26 255.255.255.192	192.168.0.0	192.168.0.1	192.168.0.62	192.168.0.63
HR	30	/27 255.255.255.224	192.168.0.64	192.168.0.65	192.168.0.94	192.168.0.95
Sales	25	/27 255.255.255.224	192.168.0.96	192.168.0.97	192.168.0.126	192.168.0.127
IT	15	/27 255.255.255.224	192.168.0.128	192.168.0.129	192.168.0.158	192.168.0.159

First unallocated address: 192.168.0.160 (addresses left in 192.168.0.0/24: 96)

Worked example 6: Test 2 2024 Q1.3, 172.16.0.0 into 8 subnets

Class B (default /16). 8 subnets need 3 bits ($2^3 = 8$), so the new prefix is **/19**, mask **255.255.224.0**, host bits 13, usable hosts $2^{13} - 2 =$ **8190**. The 8 subnets:

Subnet #	Subnet ID	Mask	First usable	Last usable	Broadcast
0	172.16.0.0/19	255.255.224.0	172.16.0.1	172.16.31.254	172.16.31.255
1	172.16.32.0/19	255.255.224.0	172.16.32.1	172.16.63.254	172.16.63.255
2	172.16.64.0/19	255.255.224.0	172.16.64.1	172.16.95.254	172.16.95.255
3	172.16.96.0/19	255.255.224.0	172.16.96.1	172.16.127.254	172.16.127.255
4	172.16.128.0/19	255.255.224.0	172.16.128.1	172.16.159.254	172.16.159.255
5	172.16.160.0/19	255.255.224.0	172.16.160.1	172.16.191.254	172.16.191.255
6	172.16.192.0/19	255.255.224.0	172.16.192.1	172.16.223.254	172.16.223.255
7	172.16.224.0/19	255.255.224.0	172.16.224.1	172.16.255.254	172.16.255.255

Subnet 3 depends on the numbering convention: counting from 0 it is **172.16.96.0/19** (first 172.16.96.1, last 172.16.127.254, broadcast 172.16.127.255); counting from 1 it is 172.16.64.0/19. State your convention. **Why subnetting prevents IP conflicts:** each subnet has its own unique network ID and non-overlapping address range, so no two devices in different subnets can be given the same address, and traffic stays inside its subnet.

Worked example 7: Exam 2024 Q1.6, MAN of 8 buildings from 164.10.0.0 (255 hosts each)

255 hosts + network + broadcast = 257 addresses, which does not fit 8 host bits (256), so each building needs **9 host bits = /23** (510 usable hosts), mask **255.255.254.0**. Second building:

Item	Value
Address / prefix	164.10.2.0/23
Subnet mask	255.255.254.0
Network ID	164.10.2.0
First usable	164.10.2.1
Last usable	164.10.3.254
Broadcast	164.10.3.255
Usable hosts	510

(Building 1 is 164.10.0.0/23.) If your lecturer's memo uses the "8 subnets" route (/19), the second building is 164.10.32.0/19; state your assumption.

Worked example 8: conversions and quick calculations

- **2023 Q1.7:** 222.1.1.20 with 255.255.255.192 = **/26**; 135.1.1.25 with 255.255.248.0 = **/21**.
- **2023 Q1.8:** /12 = **255.240.0.0**, usable $2^{20} - 2$ = **1,048,574**; /20 = **255.255.240.0**, $2^{12} - 2$ = **4,094**; /30 = **255.255.255.252**, **2** usable.
- **2019 Supp Q3.1:** 198.168.4.121/27:

Item	Value
Address / prefix	198.168.4.121/27
Subnet mask	255.255.255.224
Network ID	198.168.4.96
First usable	198.168.4.97
Last usable	198.168.4.126
Broadcast	198.168.4.127
Usable hosts	30

↳ *Answers feed [T2] ("VLSM, subnet metrics, unallocated space, Junior Technician Traps") and [D3] ("FLSM vs VLSM, CIDR"). The /32 route trick of Part A is the same prefix idea.*

PART C: The Application Layer

Source: Chapter 6/7 Application Layer deck (66 slides), starting at the **client-side and server-side steps (slide 22)**, with a short recap of the earlier slides the exam needs. Textbooks: Tanenbaum Ch. 7, Forouzan Ch. 25 to 27. **Exam tags:** [D5], [T2] port mappings.

C.0 Recap you need first (slides 2 to 18)

The application layer is the software that lets users use the network; it **uses transport services** to build distributed applications.

DNS (slides 3 to 5). Finding the IP for a hostname is **resolution**. Steps: (1) the computer asks its **local name server**; (2) the local server asks a **root name server**; (3) the root returns the name server for a lower zone; (4) continue down the zones until a server can answer. DNS runs on **UDP port 53**, retransmits lost messages, and **caches** answers. Key resource records: **A/AAAA** (IP addresses), **NS** (name servers), **MX** (mail).

Domain Resource Records (1)

The key resource records in the namespace are IP addresses (A/AAAA) and name servers (NS), but there are others too (e.g., MX)

Type	Meaning	Value
SOA	Start of authority	Parameters for this zone
A	IPv4 address of a host	32-Bit integer
AAAA	IPv6 address of a host	128-Bit integer
MX	Mail exchange	Priority, domain willing to accept email
NS	Name server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
SPF	Sender policy framework	Text encoding of mail sending policy
SRV	Service	Host that provides it
TXT	Text	Descriptive ASCII text

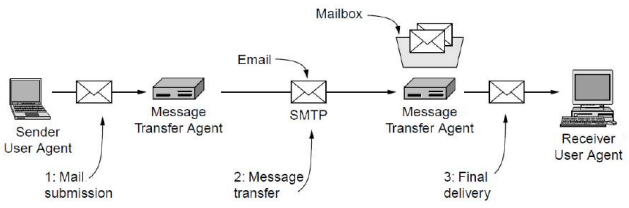
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Lecture slide 4: Domain Resource Records (1)

Email (slides 6 to 18). The user agent (what you see) hands mail to a **mail transfer agent**. **SMTP** (readable text commands) moves it: **submission** from the user agent to the MTA on **port 587**, **MTA to MTA on port 25**. Final delivery to the recipient uses **IMAP** (folders, manipulate messages) or **POP3**, or a web interface. Messages have an **envelope** (transport) and a **message** (header + body); **MIME** headers describe what is in the body (text, HTML, audio and so on).

Architecture and Services (1)

The key components and steps (numbered) to send email



Architecture of the email system

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Lecture slide 7: Architecture and Services (1)

Message Transfer (2)

Sending a message:

- From Alice to Bob
- SMTP commands are marked [pink]

```
                                S: 220 ee.uwa.edu.au SMTP service ready
C: HELO abcd.com
                                S: 250 cs.washington.edu says hello to ee.uwa.edu.au
C: MAIL FROM: <alice@cs.washington.edu>
                                S: 250 sender ok
C: RCPT TO: <bob@ee.uwa.edu.au>
                                S: 250 recipient ok
C: DATA
                                S: 354 Send mail; end with "." on a line by itself
C: From: alice@cs.washington.edu
C: To: bob@ee.uwa.edu.au
C: MIME-Version: 1.0
C: Message-Id: <0704760941.AA00747@ee.uwa.edu.au>
C: Content-Type: multipart/alternative; boundary=qwertyuiopasdfghjklzxcvbnm
C: Subject: Earth orbits sun integral number of times
C:
C: This is the preamble. The user agent ignores it. Have a nice day.
C:
C: --qwertyuiopasdfghjklzxcvbnm
C: Content-Type: text/html
C:
C: <p>Happy birthday to you
C: Happy birthday to you
   . . . (rest of message) . . .
C: --qwertyuiopasdfghjklzxcvbnm
C: .
                                S: 250 message accepted
C: QUIT
                                S: 221 ee.uwa.edu.au closing connection
```

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Lecture slide 16: Message Transfer (2)

DHCP DORA (also Part B): Discover → Offer → Request → Acknowledge gives a host an IP lease.

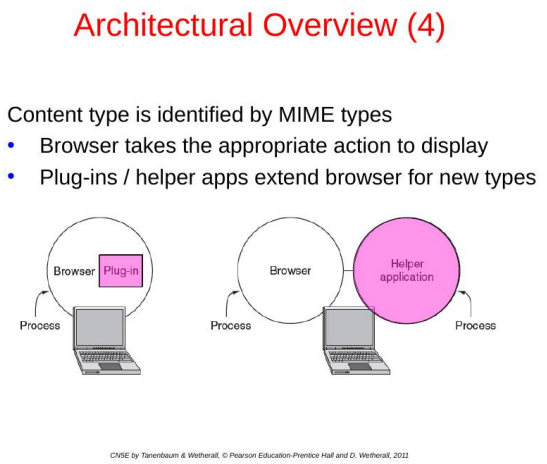
↳ *[D5] "DNS resolution, DHCP DORA process". Ports: SMTP 25, DNS 53, HTTP 80, HTTPS 443, SSH 22, FTP 21 are in Part D and the [T2] port-matching question.*

C.1 The web: client side and server side (slides 19 to 25)

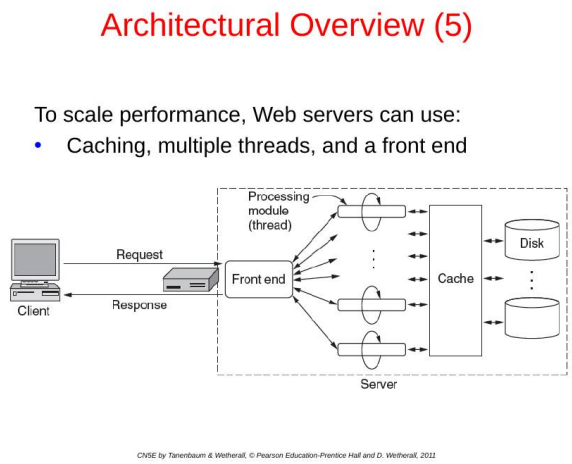
HTTP transfers pages from servers to browsers. Pages are named with **URLs**: protocol + server + page, for example http://www.phdcomics.com/comics.php (protocol http, server www.phdcomics.com, page /comics.php).

Client (browser) steps to follow a hyperlink	Server steps to serve pages
1. Determine the **protocol** (HTTP)	1. **Accept a TCP connection** from the client
2. Ask **DNS** for the server's IP address	2. Get the page request and **map it to a resource** (for example a file name)
3. Make a **TCP connection** to the server	3. **Get the resource** (file from disk)
4. **Send the request** for the page; the server sends it back	4. **Send the contents** to the client
5. **Fetch other URLs** as needed to display the page	5. **Release idle TCP connections**
6. **Close idle TCP connections**	

A fuller picture of the server (slide 25): resolve the name of the page requested; perform **access control**; **check the cache**; fetch the page from disk or **run a program**; determine the rest of the response; **return the response**; make an **entry in the server log**.



Lecture slide 23: Architectural Overview (4)



Lecture slide 24: Architectural Overview (5)

Content types (slide 23): the type is identified by **MIME types**; the browser takes the appropriate action to display it; **plug-ins / helper applications** extend the browser for new types. **Scaling a web server (slide 24): caching, multiple threads, and a front end**.

Cookies (slide 26): HTTP itself is stateless, so **cookies** support stateful interaction. The server sends cookies (state) with a page response; the client **stores them across page fetches** and **sends them back** with later requests (for example a shopping cart or login).

Architectural Overview (7)

Cookies support stateful client/server interactions

- Server sends cookies (state) with page response
- Client stores cookies across page fetches
- Client sends cookies back to server with requests

Domain	Path	Content	Expires	Secure
toms-casino.com	/	CustomerID=297793521	15-10-10 17:00	Yes
jills-store.com	/	Cart=1-00501;1-07031;2-13721	11-1-11 14:22	No
aportal.com	/	Prefs=Stk:CSCO+ORCL;Spt:Jets	31-12-20 23:59	No
sneaky.com	/	UserID=4627239101	31-12-19 23:59	No

Examples of cookies

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Lecture slide 26: Architectural Overview (7)

↳ *The client's step 2 uses DNS (C.0), step 3 uses the **TCP three-way handshake** (Part D). A client is any program that starts the conversation; a server waits for it. This is the client-server architecture from the Section 1 deck.*

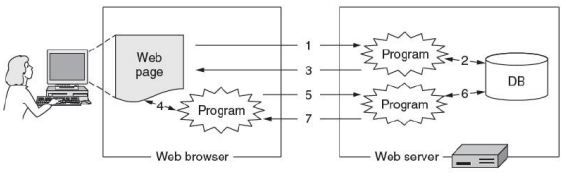
C.2 Static and dynamic pages (slides 27 to 30)

	Static page	Dynamic page / web application
What	A **file**; same contents every viewing	**Generated by a program** (with a database) and/or the client
Can it be rich?	Yes: HTML text and images, forms, style sheets, vector graphics, video	Pages vary each time, like using an application
Where code runs	none	**Server-side** (example: **PHP**) or **client-side** (example: **JavaScript**)

Dynamic Pages & Web Applications (1)

Dynamic pages are generated by programs running at the server (with a database) and the client

- E.g., PHP at server, JavaScript at client
- Pages vary each time like using an application

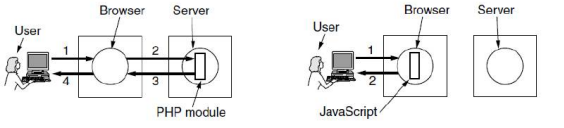


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Lecture slide 29: Dynamic Pages & Web Applications (1)

Dynamic Pages & Web Applications (2)

The difference between server and client programs



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Lecture slide 30: Dynamic Pages & Web Applications (2)

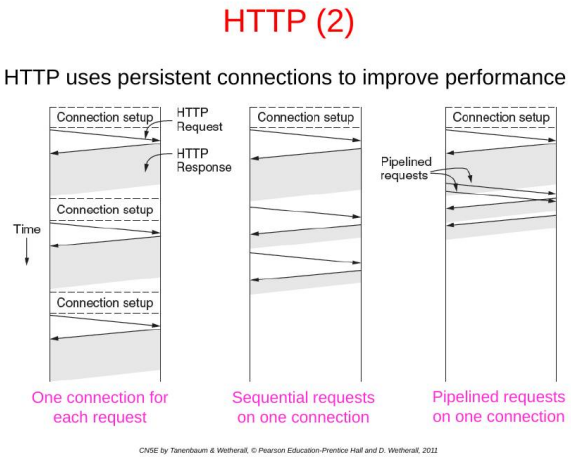
Server-side vs client-side scripting: server-side code runs on the web server before the page is sent (it can reach the database, and the user never sees it); client-side code is sent to the browser and runs there (fast interaction, no round trip to the server).

C.3 HTTP in detail (slides 31 to 35)

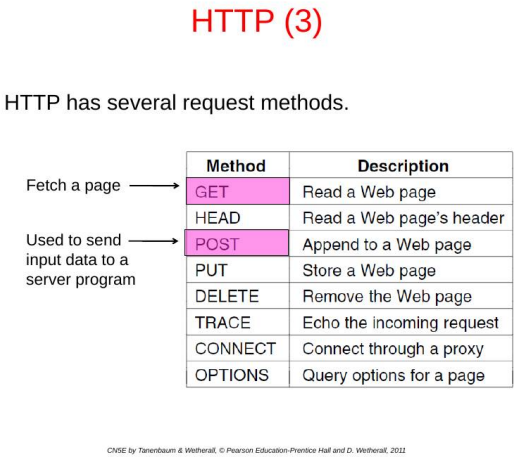
HTTP is a **request-response** protocol on top of **TCP**, server usually on **port 80**; headers are readable ASCII; content is described with MIME types; it supports **pipelining** and **caching**.

Persistent connections (slide 32) improve performance:

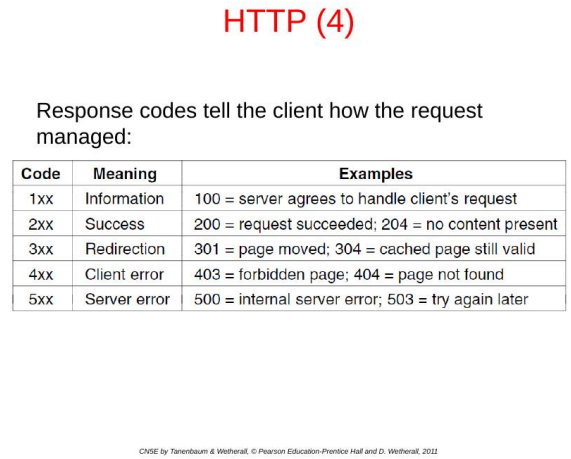
Mode	Behaviour	Result
One connection per request	Set up a TCP connection for every request	Wastes time repeatedly setting up connections
Sequential requests on one connection	Reuse the connection	No new setup each time
Pipelined requests on one connection	Send several requests without waiting	Reduces waiting time further



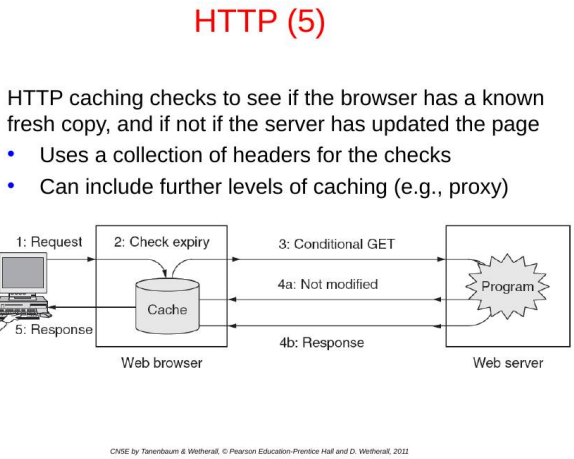
Lecture slide 32: HTTP (2)



Lecture slide 33: HTTP (3)



Lecture slide 34: HTTP (4)



Lecture slide 35: HTTP (5)

Methods (33): GET fetches a page; **POST** sends input data to a server program (and others such as HEAD, PUT, DELETE). **Response codes (34):** tell the client how the request went: 1xx information, **2xx success** (200 OK), **3xx redirection**, **4xx client error** (404 Not Found), **5xx server error**. **Caching (35):** the browser checks whether it has a **known fresh copy**, and if not whether the server has an updated page, using a set of headers; further caches (proxies) can sit in between.

↳ *HTTPS = HTTP over a secured channel on **port 443** (public-key crypto, Part A). [T2] "HTTP 80, HTTPS 443".*

C.4 The mobile web and web search (slides 36 and 37)

Mobiles are hard clients: small screens, limited input, limited bandwidth, **intermittent** connectivity (it comes and goes), limited computing power. Strategies: **content** (mobile-friendly versions, **transcoding**) and **protocols** (no need for special ones; HTTP with **header compression** is enough).

Web search needs **crawling** (follow links to new pages), and **indexing** (helped by known structure such as XML to improve relevance).

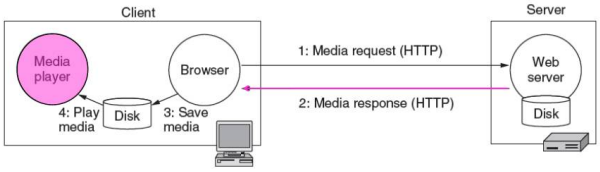
C.5 Streaming audio and video (slides 38 to 55)

Audio and video (VoIP, video streaming) are now key traffic types. **Streaming stored media** (video on demand):

Streaming Stored Media (1)

A simple method to stream stored media, e.g., for video on demand, is to fetch the video as a file download

- But has large startup delay, except for short files



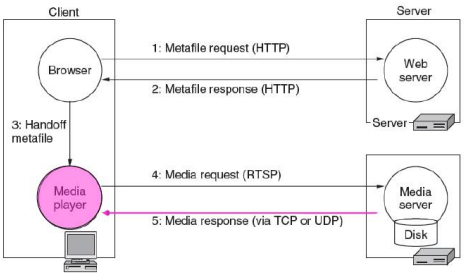
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Lecture slide 40: Streaming Stored Media (1)

Streaming Stored Media (2)

Effective streaming starts the playout during transport

- With RTSP (Real-Time Streaming Protocol)



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Lecture slide 41: Streaming Stored Media (2)

Streaming Stored Media (3)

Key problem: how to handle transmission errors

Strategy	Advantage	Disadvantage
Use reliable transport (TCP)	Repairs all errors	Increases jitter significantly
Add FEC (e.g., parity)	Repairs most errors	Increases overhead, decoding complexity and jitter
Interleave media	Masks most errors	Slightly increases decoding complexity and jitter

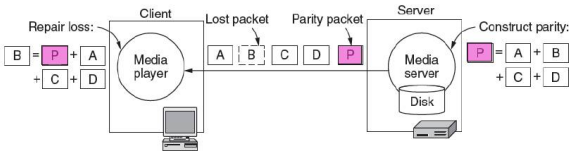
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Lecture slide 42: Streaming Stored Media (3)

Streaming Stored Media (4)

Parity packet can repair one lost packet in a group of N

- Decoding is delayed for N packets



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Lecture slide 43: Streaming Stored Media (4)

Approach	Idea	Weakness
Download as a file (slide 40)	Fetch the whole file first	Large **start-up delay**
Streaming (41)	**Start playout during transport**, using **RTSP** (Real-Time Streaming Protocol)	Must handle errors
Three server designs (44 to 46)	(1) web server; (2) web server plus a **metafile**; (3) a dedicated **media server**	More control as you go down

Handling transmission errors (slide 42):

Strategy	Advantage	Disadvantage
Use reliable transport (**TCP**)	Repairs all errors	Increases **jitter** a lot
Add **FEC** (for example parity)	Repairs most errors	Overhead, decoding complexity, jitter
Interleave media	Masks most errors	Slightly more decoding complexity and jitter

Parity packet (43): one extra parity packet can repair **one lost packet in a group of N**; decoding is delayed for N packets. (FEC = Forward Error Correction; same idea as Part E.)

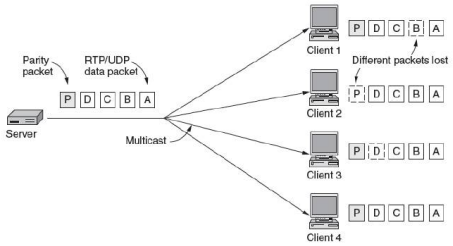
Streaming live media (47 to 50): like the stored case, but you **cannot stream faster than the live rate**, and many users watch at once. **UDP with multicast** is efficient but rarely available, so many TCP connections are used. With multicast, **parity works well**: each client can lose a different packet and recover. Examples: internet radio, ITV, IPTV; protocols such as **UDP and RTP**.

Real-time conferencing (51 to 55): two or more connected live streams (VoIP, Skype). The key issue is **low (interactive) latency**; it benefits from **QoS** or ample bandwidth.

Streaming Live Media (2)

With multicast streaming media, parity is effective

- Clients can each lose a different packet and recover

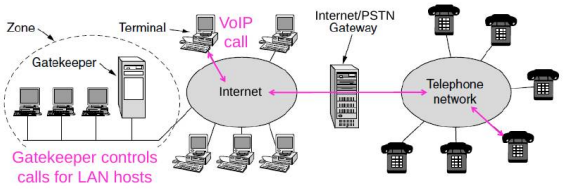


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Lecture slide 48: Streaming Live Media (2)

Real-Time Conferencing (2)

H.323 architecture for Internet telephony supports calls between Internet computers and PSTN phones.



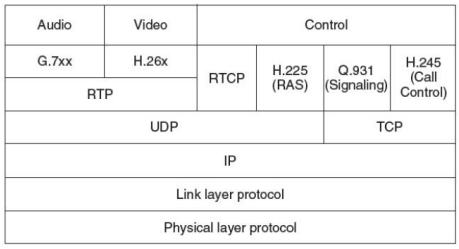
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Lecture slide 52: Real-Time Conferencing (2)

Real-Time Conferencing (3)

H.323 protocol stack:

- Call is digital audio/video over RTP/UDP/IP
- Call setup is handled by other protocols (Q.931 etc.)



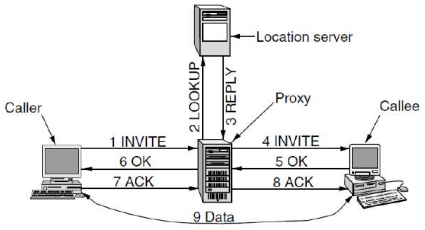
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Lecture slide 53: Real-Time Conferencing (3)

Real-Time Conferencing (6)

Setting up a call with the SIP three-way handshake

- Proxy server lets a remote callee be connected
- Call data flows directly between caller/callee



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Lecture slide 55: Real-Time Conferencing (6)

Protocol	Key points
H.323	Architecture for Internet telephony: calls between Internet computers and PSTN phones; a **gatekeeper** controls calls for LAN hosts; the call is **digital audio/video over RTP/UDP/IP**; setup by other protocols (Q.931 and so on)
SIP (Session Initiation Protocol)	Alternative to H.323; **simple, text-based**, with **URLs** as addresses; data carried with **RTP/RTCP**; call set up with a **SIP three-way handshake**; a proxy server connects a remote callee, then **call data flows directly** between caller and callee

↳ *Why UDP/RTP for live traffic: retransmitting a late voice packet is useless, so the reliability of TCP (Part D) is traded for lower delay. QoS ties back to Part B.8.*

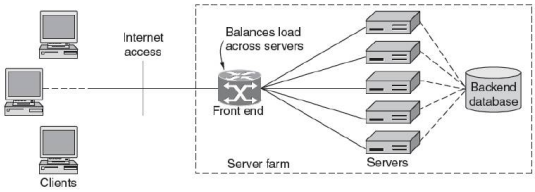
C.6 Content delivery (slides 56 to 65)

Content (web, video) is a major share of Internet traffic. Traffic has **many small/unpopular and a few large/popular flows** ("mice and elephants").

Server Farms and Web Proxies (1)

Server farms enable large-scale Web servers:

- Front-end load-balances requests over servers
- Servers access the same backend database



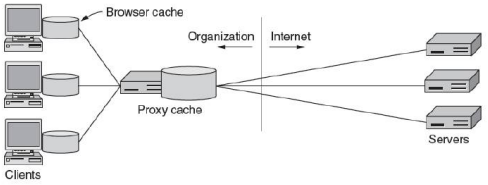
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Lecture slide 59: Server Farms and Web Proxies (1)

Server Farms and Web Proxies (2)

Proxy caches help organizations to scale the Web

- Caches server content over clients for performance
- Also implements organization policies (e.g., access)

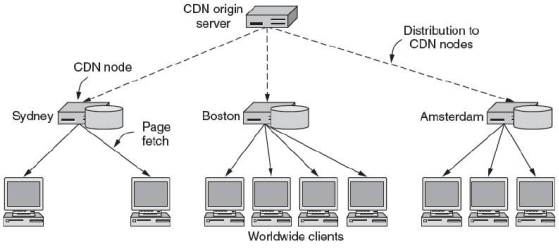


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Lecture slide 60: Server Farms and Web Proxies (2)

CDNs – Content Delivery Networks (1)

CDNs scale Web servers by having clients get content from a nearby CDN node (cache)



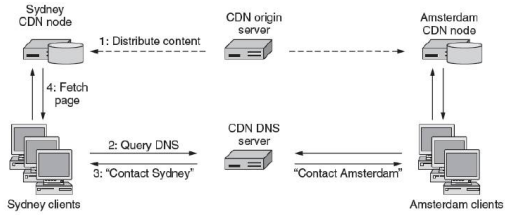
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Lecture slide 61: CDNs – Content Delivery Networks (1)

Content Delivery Networks (2)

Directing clients to nearby CDN nodes with DNS:

- Client query returns local CDN node as response
- Local CDN node caches content for nearby clients and reduces load on the origin server



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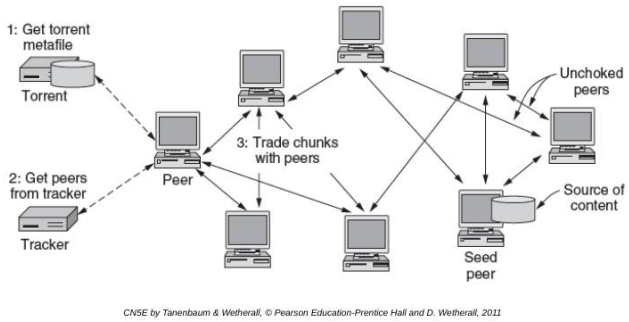
Lecture slide 62: Content Delivery Networks (2)

Technique	How it scales
Server farms (59)	A **front end load-balances** requests over many servers; all access the **same back-end database**
Web proxies (60)	A **proxy cache** keeps server content close to clients for performance, and applies organisation policy (for example access control)
CDNs (61, 62)	Clients get content from a **nearby CDN node (cache)**; **DNS directs** the client to the nearby node, which caches content for local clients and **reduces load on the origin server**
Peer-to-peer (63 to 65)	No dedicated servers: clients serve each other. Challenges: how do peers **find** each other, **download quickly**, and **encourage uploading**? **BitTorrent**: peers find each other via a **tracker** in the torrent file, swap **chunks**, preferring partners who send fastest. **DHT** = a fully distributed index (follow an O(log N) path) that can replace the tracker

Peer-to-Peer Networks (2)

Bit Torrent lets peers download torrents

- Peers find each other via Tracker in torrent file
- Peers swap chunks (parts of content) with partners, preferring those who send most quickly [2]
- Many peers speed download; preference helps uploads [3]



Lecture slide 64: Peer-to-Peer Networks (2)

↳ *CDN request routing uses **DNS** (C.0). A cache that serves nearby clients is the same idea as the HTTP cache (C.3).*

C.7 Exam-ready answers for Part C

Describe what happens when a browser loads a page (client + server). Browser reads the URL and picks HTTP; asks DNS for the server IP (UDP 53); opens a TCP connection (three-way handshake) to port 80; sends the GET request; the server maps it to a resource, checks access control and cache, fetches or generates the page, returns the response, logs it; the browser fetches the other URLs; idle connections are closed. **Match services to ports ([T2] 2026 Q2.2).** Web browser viewing course info → **HTTP, port 80**; uploading research datasets to a file server → **FTP, port 21**; sending official email → **SMTP, port 25**. (Spare ports in the list: SSH 22, DNS 53.) **DNS resolution ([D5]).** Local name server → root → lower zone server → ... until the answer; UDP 53; answers cached.

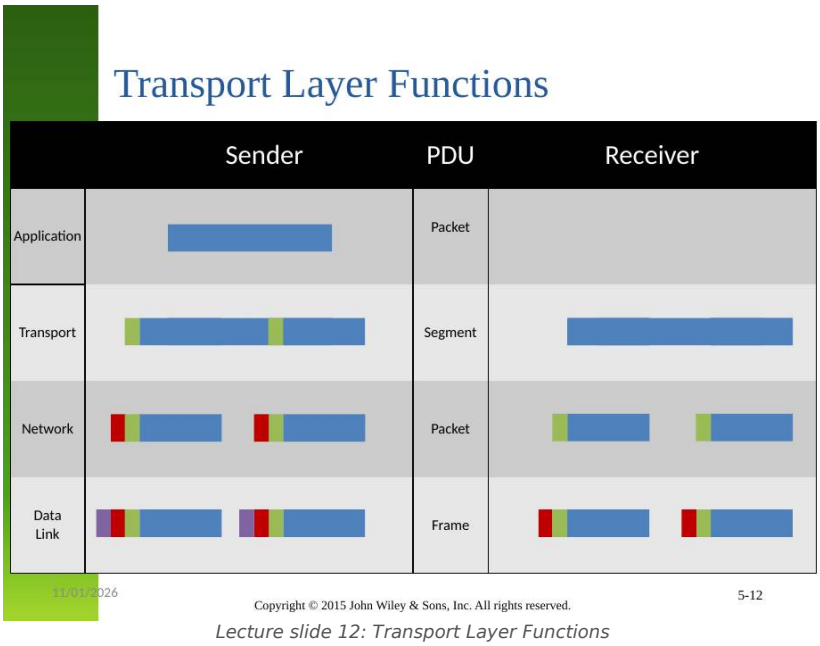
PART D: The Transport Layer

Source: Chapter 5 Transport Layer deck, slides 6 to 27 (slides 29 onward repeat the Application Layer deck). Textbooks: Tanenbaum Ch. 6, Forouzan Ch. 23 to 24. **Exam tags:** [T2] (TCP vs UDP, handshake math, ports), [D4].

D.1 What the transport layer does (slides 6 and 11)

It is **Layer 4**: it links the application and network layers and is responsible for **end-to-end delivery of messages**: **segmentation and reassembly**, and **session management**. Segmenting means breaking large files into smaller segments and putting them back together.

Layer	PDU (data unit)	What is added
Application	Data / message	
Transport	**Segment**	TCP/UDP header (ports and so on)
Network	**Packet**	IP header (IP addresses)
Data link	**Frame**	Source/destination **MAC addresses** (and trailer)



Lecture slide 12: Transport Layer Functions

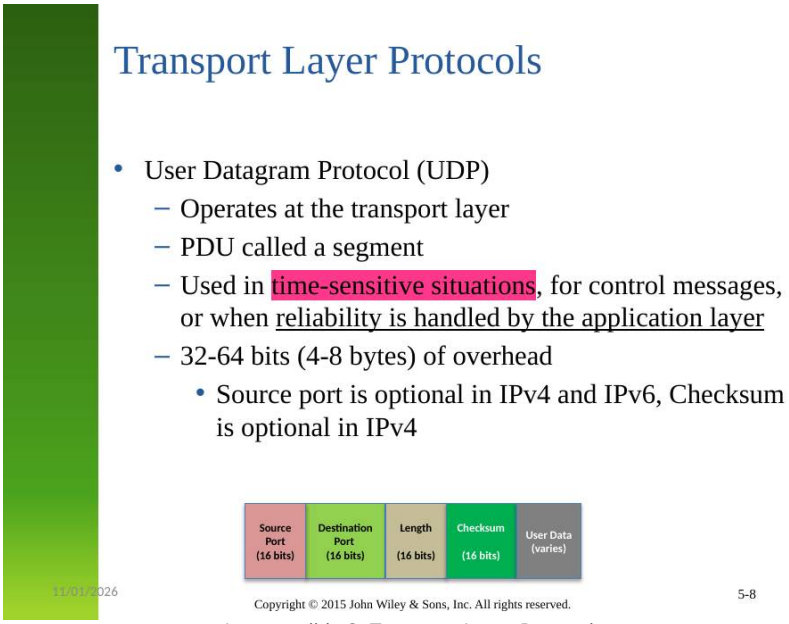
This "wrapping at each layer" is **encapsulation**.

↳ *Encapsulation is Exam 2024 Q1.1 ("first four OSI layers and encapsulation"). Tunnelling in Parts A and B is encapsulation too.*

D.2 TCP and UDP (slides 7 and 8)

	TCP	UDP
Type	**Connection-oriented** (session first)	**Connectionless** (no session)
Reliability	**Reliable**: delivered correctly and **in order**	**No** retransmission, delivery or ordering guarantee
Overhead	Larger header	Only **8 bytes** (32 to 64 bits)
Used for	Web, email, file transfer	**Time-sensitive** traffic, control messages, or when the **application handles reliability** (DNS, streaming, VoIP)
Mechanisms	Handshake, sequence numbers, ACKs, windows	Checksum (optional in IPv4)

UDP header (4 fields, 16 bits each): Source port (optional in IPv4/IPv6), Destination port, Length, Checksum, then user data.



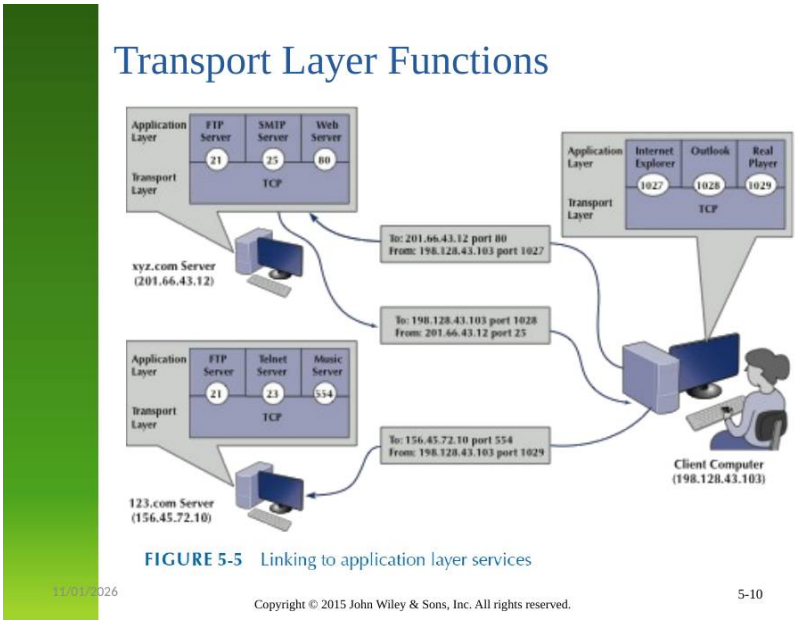
Lecture slide 8: Transport Layer Protocols

↳ *[T2] "TCP vs UDP state/reliability trade-offs". Streaming and DNS using UDP: Part C. TCP "state" is exactly what a SYN flood abuses (Part A).*

D.3 Ports (slide 9)

A host runs many applications, so **ports** (2-byte numbers) identify which application a segment is for. The pair **IP address + port** is a **socket**; a connection is identified by the sockets at both ends.

Service	Protocol and port
HTTP (web)	TCP **80**
HTTPS (secure web)	TCP **443**
FTP	TCP **20 and 21** (21 = control)
SSH (secure remote login)	TCP **22**
SMTP (sending email)	TCP **25**
DNS	**UDP 53** most commonly (also TCP 53)
POP3	TCP 110 (secure version 995)
IMAP	TCP 143



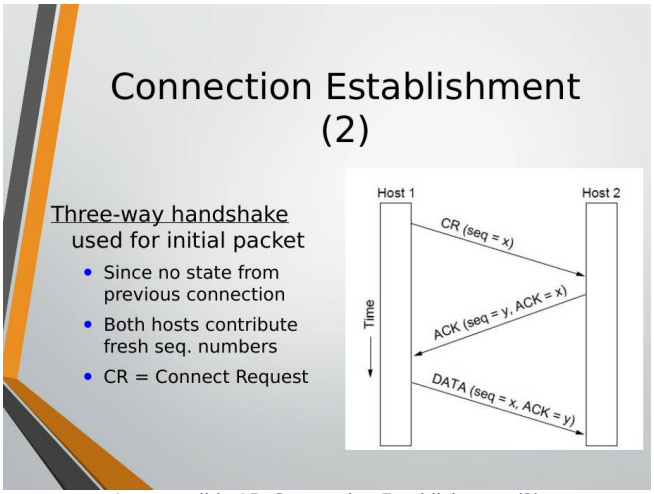
Lecture slide 10: Transport Layer Functions

↳ *Memorise the first six for [T2]/[D4]. NAT (Part B.10) uses ports to tell connections apart; Mobile IP registration uses UDP 434 (Part A).*

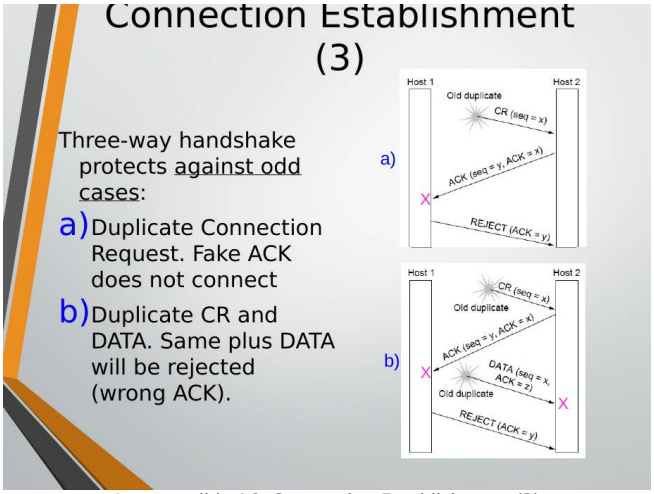
D.4 Session management and the three-way handshake (slides 13 to 16)

A **session** (virtual circuit, a conversation) is **connection-oriented** messaging (TCP). Sending without one is **connectionless** (UDP). TCP opens with a **three-way handshake**: **SYN → SYN-ACK → ACK**.

Why three ways? Packets can be lost, corrupted, delayed and duplicated, so an old or duplicate connection request must not be treated as new. Each side contributes a **fresh initial sequence number (ISN)**: the first host picks x, the second picks y. Sequence numbers are not reused within twice the **maximum segment lifetime (MSL)**: **2T = 240 s** (slide 14). Errors are handled with **ARQ and checksums**.



Lecture slide 15: Connection Establishment (2)



Lecture slide 16: Connection Establishment (3)

Protection against odd cases (slide 16): a **duplicate connection request** arrives; the second host answers, but the first host **rejects** it because the numbers do not match a current connection (a **fake ACK does not connect**). **Duplicate request plus data**: the data is rejected as well (wrong ACK).

The sequence/ACK arithmetic (the exam calculation)

Rule: **a SYN uses up one sequence number**, so the ACK number is always **the other side's ISN + 1**.

Segment	Flags	Sequence number	ACK number
1. A → B	SYN	ISN_A	n/a
2. B → A	SYN-ACK	ISN_B	**ISN_A + 1**
3. A → B	ACK	**ISN_A + 1**	**ISN_B + 1**

Test 2 2026 Q2.1 (ISN_A = 1,543, ISN_B = 2,780)

Segment	Flags	Sequence number	ACK number
1. Host A → Host B	SYN	1,543	n/a
2. Host B → Host A	SYN-ACK	2,780	1,543 + 1 = **1,544**
3. Host A → Host B	ACK	1,543 + 1 = **1,544**	2,780 + 1 = **2,781**

Special Test 2 Q2 (ISN_Student = 4,250, ISN_Server = 7,600)

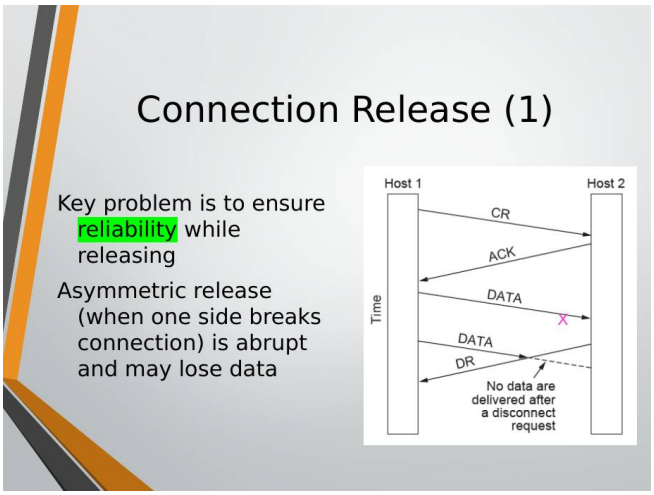
Segment	Flags	Sequence number	ACK number
1. Student → Server	SYN	4,250	n/a
2. Server → Student	SYN-ACK	7,600	4,250 + 1 = **4,251**
3. Student → Server	ACK	4,250 + 1 = **4,251**	7,600 + 1 = **7,601**

Drawing it (5 marks): two vertical timelines, Host A on the left and Host B on the right, time downward. Arrow 1 A→B labelled "SYN, seq = ISN_A"; arrow 2 B→A labelled "SYN-ACK, seq = ISN_B, ack = ISN_A + 1"; arrow 3 A→B labelled "ACK, seq = ISN_A + 1, ack = ISN_B + 1". Mark the states: CLOSED → SYN-SENT → ESTABLISHED on A; LISTEN → SYN-RECEIVED → ESTABLISHED on B.

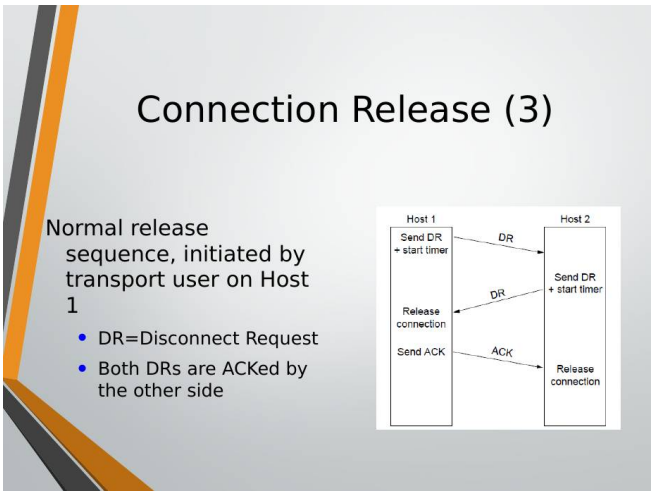
↳ *A **SYN flood** (Part A) sends step 1 and never step 3. The ISN idea is why a late duplicate cannot hijack a new connection.*

D.5 Connection release (slides 17 to 19)

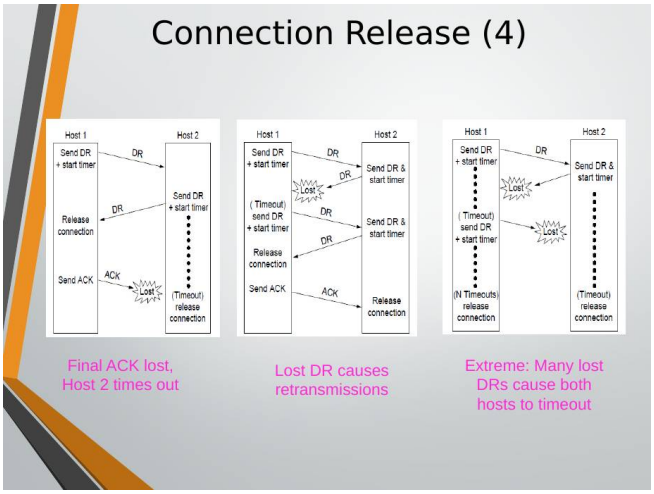
Release must be reliable too. **Asymmetric release** (one side just breaks the connection) is abrupt and may **lose data**. **Normal (symmetric) release (slide 18):** Host 1 sends a **DR (Disconnect Request)** and starts a timer; Host 2 replies with its own DR; Host 1 ACKs and releases; when Host 2 gets the ACK it releases too. Both DRs are ACKed.



Lecture slide 17: Connection Release (1)



Lecture slide 18: Connection Release (3)



Lecture slide 19: Connection Release (4)

When messages are lost (slide 19): (left) the **final ACK is lost**, so Host 2 times out and releases anyway; (middle) a **DR is lost**, so Host 1 times out and **retransmits**; (right) many DRs are lost, so after **N timeouts** Host 1 gives up and releases, and Host 2 eventually times out. **The timer prevents anyone waiting forever.**

D.6 Error control and flow control (slide 21)

	Error control	Flow control
Goal	Detect and **retransmit** lost or corrupted data	Make sure the sender does **not send faster than the receiver can process**
How	**Sliding window** (from the link layer) + **checksums** + **retransmissions**	The **window** tells the sender how much **buffer space** the receiver has; a **variable-size sliding window** grows or shrinks with receiver space

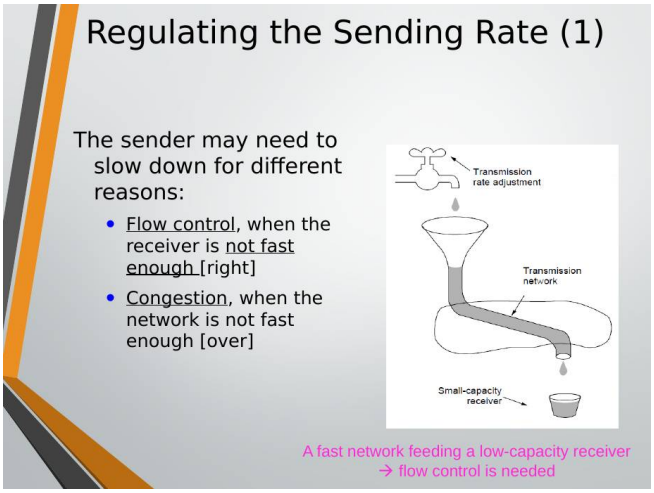
A **checksum** is a value calculated from the data before sending and checked on arrival (Part E).

> ↳ *The same sliding window and ARQ appear in the Data link part (E.8). A fast sender into a slow receiver is the slide-23 picture: "a fast network feeding a low-capacity receiver needs flow control".*

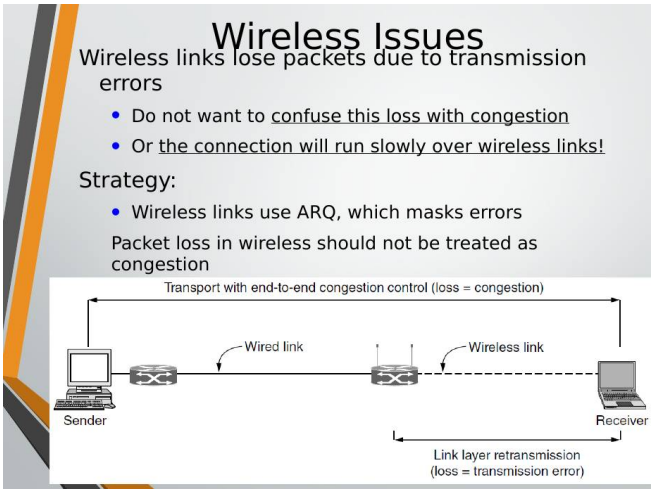
D.7 Congestion control (slides 22 to 24)

Two layers share the job: the **transport layer controls the offered load**; the **network layer is where congestion happens** (Part B.7). Analogy: the transport layer controls how many cars enter the road; the network layer is the road that jams.

Issue	Meaning
Desirable bandwidth allocation	Share bandwidth **fairly and efficiently** among users
Regulating the sending rate	Slow down when the receiver (**flow control**) or the network (**congestion**) is not fast enough
Wireless issues	Wireless links lose packets to **transmission errors** (interference, weak signal); this must **not** be read as congestion or the connection runs slowly. Strategy: the wireless link uses **ARQ** to mask errors



Lecture slide 23: Regulating the Sending Rate (1)



Lecture slide 24: Wireless Issues

↳ ***Flow control vs congestion control*** *is named in [D4] and Part B.7. The wireless-loss problem is the same TCP weakness as Part A.6.*

D.8 Performance issues (slides 26 and 27)

Topic	Idea
Performance problems	Congestion, too many small packets, excessive traffic
Measuring performance	**Throughput, delay, response time**
Host design for fast networks	Design sender/receiver to process data fast
Fast segment processing	Process incoming/outgoing segments quickly
Header compression	Shrink headers so less bandwidth is spent on control information
"Long fat" networks	High bandwidth but **long delay** (long-distance high-speed links) need special protocols

Unexpected loads cause problems: broadcast storm (one broadcast triggers another until the network is overwhelmed); **synchronization** (500 students enter the office at once: after a power failure every computer contacts the DHCP server together); **tiny packets** (TCP sends many small packets instead of a few large ones).

↳ *2016 Q1.1:* ***throughput*** *and the factors that determine it (see F.7).*

D.9 Exam-ready answers for Part D

TCP vs UDP. TCP: connection-oriented, reliable, ordered, flow and congestion control, handshake, bigger header, state kept at both ends. UDP: connectionless, no guarantees, 8-byte header, no setup delay; chosen for real-time traffic, control messages, or when the application provides reliability. **Handshake numbers.** Always: SYN-ACK acks **ISN_A + 1**; final ACK has seq **ISN_A + 1** and ack **ISN_B + 1**. Show the tables above for your numbers. **Why does the handshake use fresh sequence numbers from both sides?** So duplicate/old connection requests and data cannot be mistaken for a new connection.

PART E: The Data Link Layer and Medium Access Protocols

Source: Datalink layer deck (59 slides) and Medium Access Protocols (continuation) deck (12 slides). Textbooks: Tanenbaum Ch. 3 to 4, Forouzan Ch. 9 to 14. **Exam tags:** [T1], [T2], [D2].

E.1 What the data link layer does (slides 2 and 3)

Layer 2. It moves messages **from one device to the next on the same link**, controls how messages are sent on the media, and **organises the Layer 1 bit stream into coherent messages (frames)** for the network layer. Three major functions:

Function	Question it answers
Media Access Control (MAC)	Which device may transmit, and when?
Error control	Did the frame arrive correctly, and how do we recover if not?
Message delineation	Where does a frame start and end?

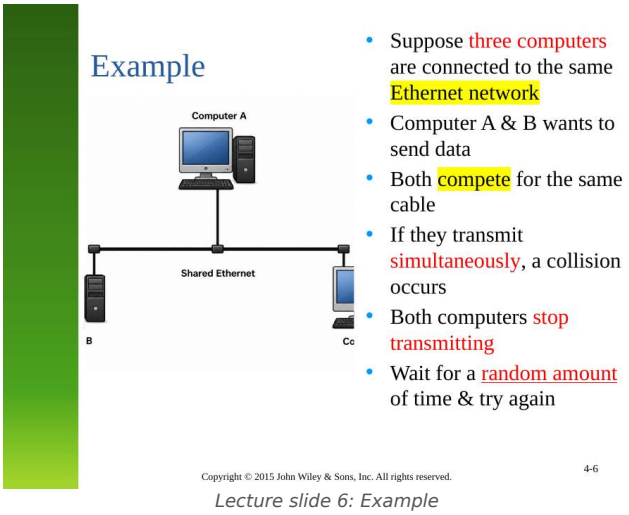
↳ *Layer 2 delivers by **MAC address inside one subnet**; leaving the subnet needs Layer 3 (Part B). This is the [T1] OSI-troubleshooting split. Layer 1 (Part F) gives it raw bits.*

E.2 Media access control: two approaches (slides 4 to 13)

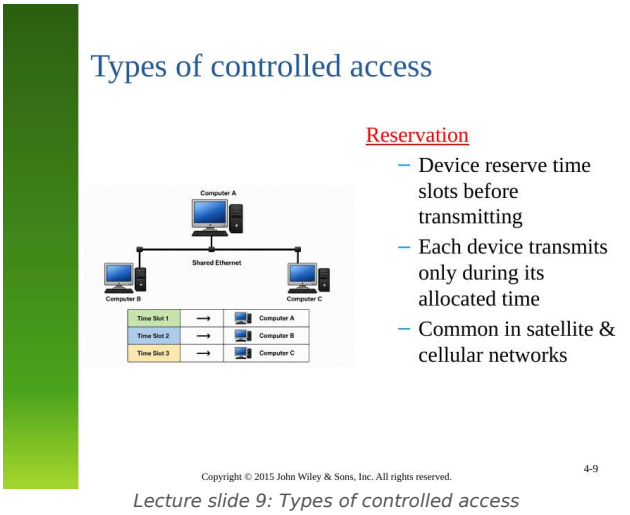
MAC matters on **multipoint (shared) circuits** and half-duplex point-to-point circuits.

	Contention access	Controlled access
Idea	Devices **transmit whenever the circuit is free**; no central control	The network decides **who may send and when**
Collisions	Possible (devices compete)	Greatly reduced or avoided
Example	**Ethernet** LANs, Wi-Fi	**Reservation, polling, token passing**
Analogy	5 people in a meeting all speaking at once	The chair says who speaks next

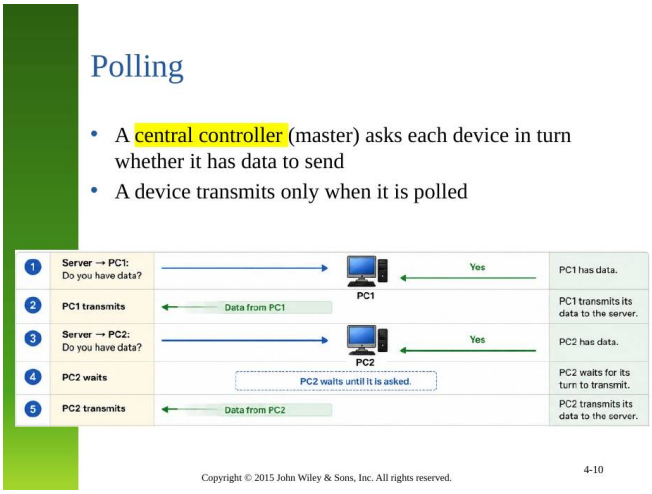
Contention example (slide 6): computers A and B both want to send on the same Ethernet cable; if they transmit at once a **collision** occurs; both stop, wait a **random** time and try again. **In Wi-Fi (slide 7):** devices listen first; if busy they wait a random period.



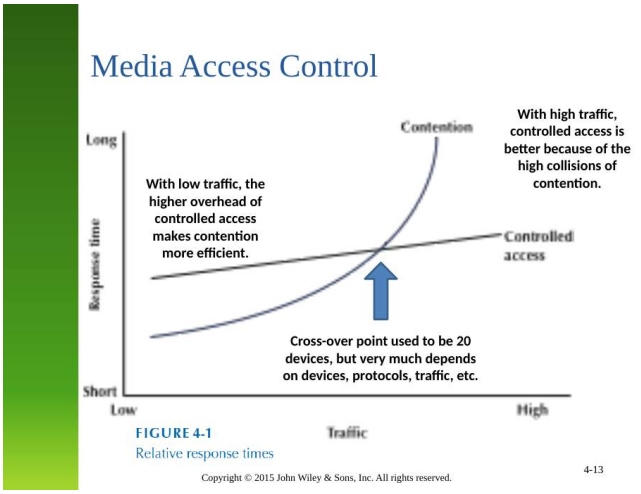
Lecture slide 6: Example



Lecture slide 9: Types of controlled access



Lecture slide 10: Polling



Lecture slide 13: Media Access Control

Controlled access types (slides 9 to 11):

Type	How it works	Used in
Reservation	Devices **reserve time slots** first; each transmits only in its slot	Satellite and cellular networks
Polling	A **central controller (master) asks each device in turn** if it has data; a device sends only when polled	Master/slave setups
Token passing	A **token** circulates; **only the device holding it may send** (A → token → B → token → C)	Ring-style networks
Advantages (slide 12)	Drawbacks	
Prevents/greatly reduces collisions	More complex to implement	
Fair access to all devices	**Delays** while waiting for permission or the token	
More efficient under heavy traffic	Failure of the controller/token can affect everyone	

Which is better? (slide 13): with **high traffic**, controlled access wins (contention suffers many collisions); with **low traffic**, contention wins (controlled access has more overhead). The old cross-over was about 20 devices, but it depends on devices, protocols and traffic.

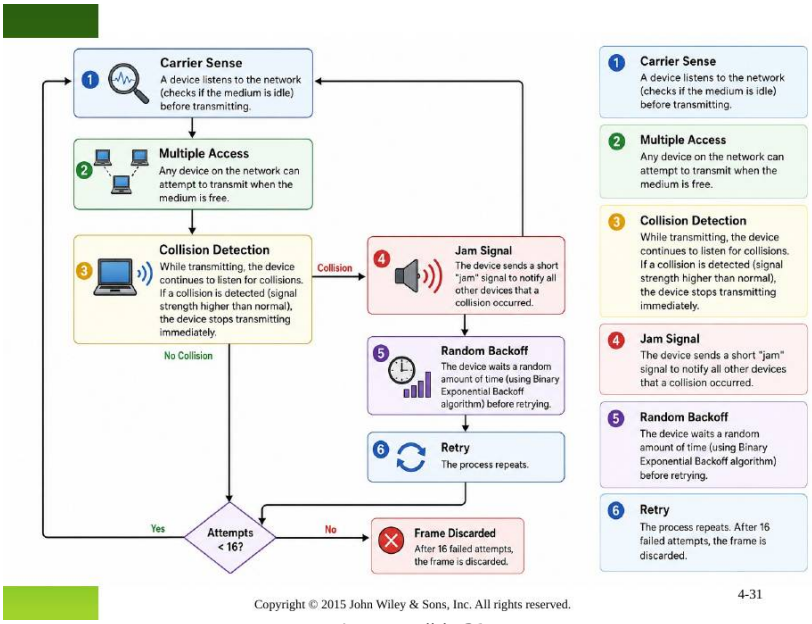
↳ *[D2] "Polling" and Exam 2024 Q2.3 (evaluate ALOHA vs polling). Reservation and cellular link to handoff in Part A.*

E.3 CSMA and CSMA/CD (slides 14, 15 and 28 to 31)

CSMA (Carrier Sense Multiple Access): each host **listens before transmitting**. Busy → wait until finished. Idle → send. After a frame the network observes an **interframe gap (IFG)**. This reduces, but does not remove, collisions. If no ACK arrives, assume a collision and retransmit.

CSMA/CD (Collision Detection) is the standard access method for **half-duplex Ethernet** (10 and 100 Mbps). In early bus/hub Ethernet all devices shared one cable, so simultaneous sends collided. CSMA/CD gives rules to **listen before talking, detect collisions, and recover efficiently**.

Step	Action
1	**Listen**: wait until the channel is idle, then send
2	**Detect** a collision while sending
3	Send a **jam signal** (32 bit times) so every node knows
4	**Exponential backoff** in multiples of **512 bit times** (longer than a round-trip time, so a station that waits longer will notice the earlier retransmission before starting)
5	Retry



Lecture slide 31

↳ 2016 Q1.3 and Exam 2024 Q2.1 ask for CSMA/CD operation and its effect on TCP/IP applications (more traffic → more collisions → more retransmissions and delay). Hubs flood to every port (shared medium, [T1] Q4); switches learn MAC addresses.

E.4 Wireless: CSMA/CA, the hidden terminal and RTS/CTS (slides 16 to 27)

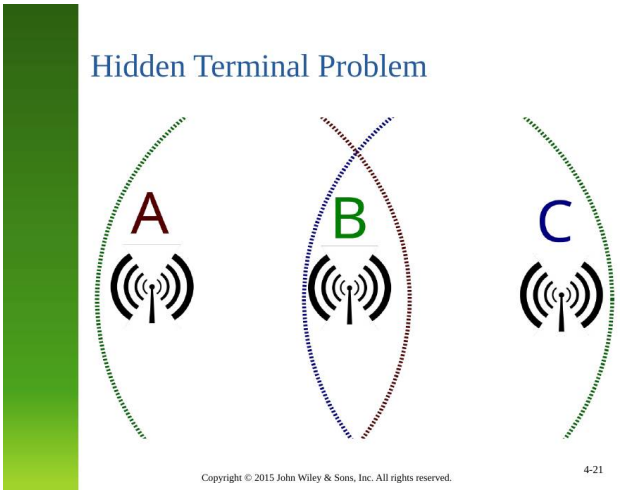
CSMA/CA (Collision Avoidance) is the medium access protocol of **IEEE 802.11 Wi-Fi**.

Step	What happens
1. **Carrier sense**	The device listens to see whether the channel is idle or busy; if busy, wait until free
2. **Random backoff**	When idle, wait a **random period** so devices do not all start at the same instant
3. **Transmit**	When the backoff timer expires, send the frame
4. **ACK**	The receiver sends an **ACK**; **no ACK = failure**, so the sender retransmits

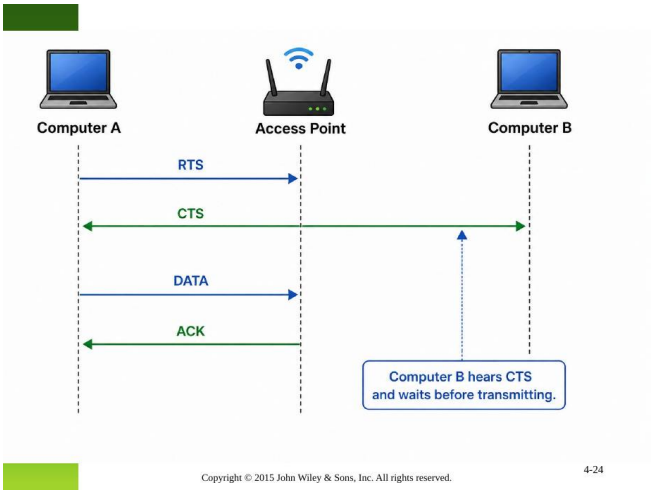
Why Wi-Fi avoids instead of detecting collisions (slides 19, 25 to 27): a wireless device **cannot transmit and listen at the same time** (half-duplex radios), and **its own transmitted signal is much stronger than any incoming signal**, so weak incoming signals are masked and collisions cannot be reliably detected. So Wi-Fi **avoids** collisions instead of detecting them. Other wireless challenges (slide 25):

Challenge	Meaning
No fate sharing	Different devices see different channel conditions and reliability
High loss rates	Interference, fading and obstacles lose packets
Variable channel conditions	Signal quality changes with movement and interference
Half-duplex radios	Cannot transmit and receive on the same channel at once
Strong transmit signal	Own signal overwhelms incoming signals, preventing collision detection

The hidden terminal problem (slides 20 to 22). Two devices cannot hear each other (out of range, or a wall between them) but **both can reach the same access point**. A and B each sense a free channel, both transmit, and their frames **collide at the access point**. Cause: limited transmission range; obstacles block signals; a device only senses transmissions within its own range.



Lecture slide 21: Hidden Terminal Problem



Lecture slide 24

Solution: RTS/CTS (slide 23). (1) The sender sends an **RTS (Request to Send)** to the access point. (2) If the channel is available, the AP replies with a **CTS (Clear to Send)**. (3) **All nearby devices that hear the CTS know the medium will be busy and wait.** (4) The sender transmits the data. (5) The receiver sends an **ACK**. Each listening station keeps an internal timer, the **NAV (Network Allocation Vector)**, set from the duration announced in RTS/CTS. A hidden station cannot hear the RTS but **hears the AP's CTS**, which is the whole point. If two RTS frames collide, no CTS comes back; the sender backs off and retries.

	CSMA/CD	CSMA/CA
Medium	Wired Ethernet	Wireless 802.11
Listens while sending?	Yes	**No** (half duplex)
Collision handling	Detect, abort, retry	**Cannot detect**; inferred from a missing ACK
ACKs needed?	No	**Yes**
Random backoff before first send?	No	**Yes**
Hidden terminal fix	Not needed	**RTS/CTS + NAV**

[T1] 2026 Q5d "Student Trap": why not use CSMA/CD on Wi-Fi? (1) radios are half duplex and cannot listen while transmitting; (2) hidden terminals mean the collision happens at the receiver, out of the sender's hearing; (3) the own signal drowns the weak incoming one; (4) detecting a collision after the whole frame has been sent wastes airtime. Wi-Fi must avoid collisions and use ACKs.

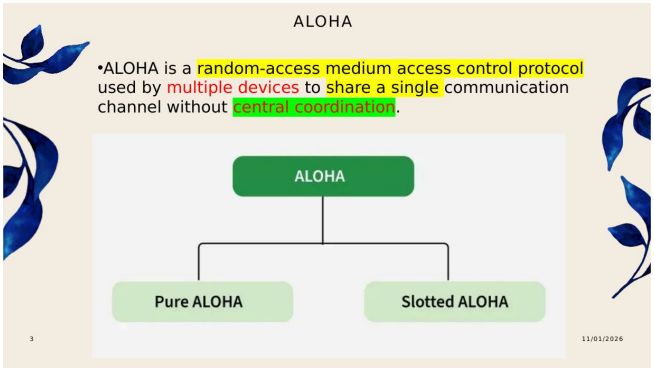
↳ *The same story from the mobile side: Part A (Mobile IP) sits above this layer. [T1] Q5b, [T2] 2024 Q1.1 (11 marks), Exam 2024 Q2.2 (4 marks, diagram) all test this.*

E.5 ALOHA (Medium Access Protocols deck, slides 3 to 12)

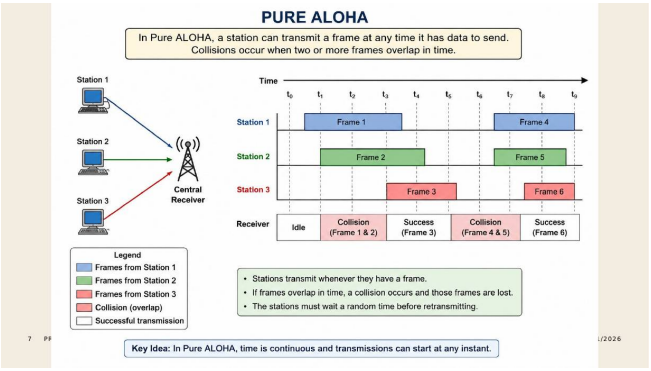
ALOHA was the **first MAC protocol**: a random-access protocol for devices sharing **one channel without central coordination**. It started as users sharing a channel to a central computer.

	Pure ALOHA	Slotted ALOHA
Time	**Continuous**; no slots	Divided into **equal slots**, each long enough for one frame
When to send	**Any time**	Only at the **start of a slot**
Carrier sensing	**None**	None (needs **time synchronisation**)
Collision	Overlapping transmissions; frames are corrupted and lost	Two or more stations in the **same slot**
Efficiency	Low	Better (about double): fewer collisions, higher utilisation
Drawbacks	Many collisions, **low channel efficiency**, frequent retransmissions, delay, poor under heavy traffic, bandwidth wasted	All stations must agree on slot timing; collisions still happen; a station **waits for the next slot**

How pure ALOHA works: (1) any station transmits any time; (2) no carrier sensing; (3) overlapping frames collide and are corrupted; (4) the receiver sends an **ACK** if the frame arrived; no ACK within a **timeout** means a collision; (5) the sender waits a **random backoff** and retransmits until success or a maximum number of attempts.



Lecture slide 3: ALOHA



Lecture slide 7

(Extra, standard theory: maximum efficiency is about **18%** for pure ALOHA and **37%** for slotted ALOHA; slotting halves the vulnerable period. Check against your lecturer's notes.)

> **Exam 2024 Q2.3 (5 marks): ALOHA vs polling for a new office LAN.** ALOHA: simple, no coordination, fine for light, bursty traffic, but collisions grow quickly with load. Polling: controller grants turns, no collisions, predictable delay and fairness, better under heavy load, but needs a controller (single point of failure) and adds polling delay. For a new office LAN with many users, polling (or CSMA-style access) is the better choice.

↳ *[D2] "Pure/Slotted ALOHA, TDMA, Polling". Slotted ALOHA's time slots are the same idea as TDM in Part F.*

E.6 Error control: sources and prevention (slides 32 to 37)

Error control = detecting and correcting errors so data arrives **accurately, reliably and without corruption**; typically by **retransmitting unacknowledged frames**, with a **timer** protecting against lost ACKs.

Source of error	Cause	Common effect
Line noise	Unwanted electrical signals	Bit errors
Attenuation	Signal weakens with distance	Data loss
Distortion	Signal changes shape or timing	Incorrect data
Crosstalk	Interference from nearby cables	Corrupted data
EMI	Interference from electrical devices	Signal degradation
RFI	Interference from radio signals	Communication errors
Faulty hardware	Damaged cables or devices	Lost/corrupted packets
Congestion	Excessive traffic	Delay and loss
Collisions	Simultaneous transmissions	Corrupted frames
Wireless interference	Obstacles/competing signals	Weak signal, retransmissions

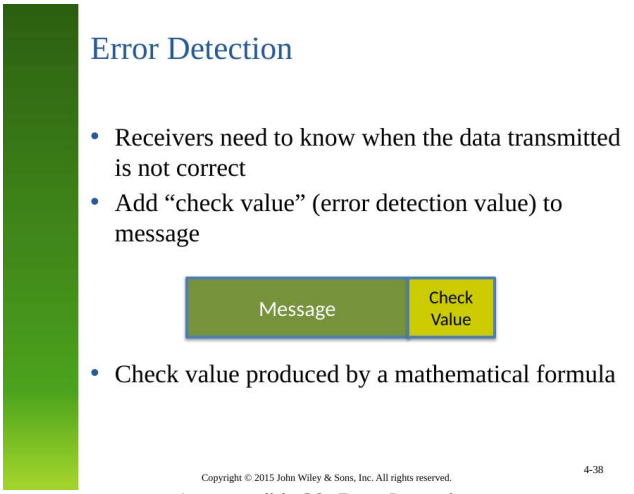
Error prevention reduces errors **before they occur** (less noise, interference, attenuation); it improves reliability, cuts retransmissions and keeps data intact.

Error	What causes it	How to prevent or fix
White noise	Movement of electrons	Increase signal strength
Impulse noise	Sudden surges (for example **lightning**)	Shield or move the wires
Cross-talk	Guard bands too small or wires too close	Increase guard bands, move or shield wires
Echo	Poor (misaligned) connections	Fix connections, tune equipment
Attenuation	Gradual signal loss with distance	Use **repeaters**
Intermodulation noise	Signals from several circuits combine	Move or shield the wires

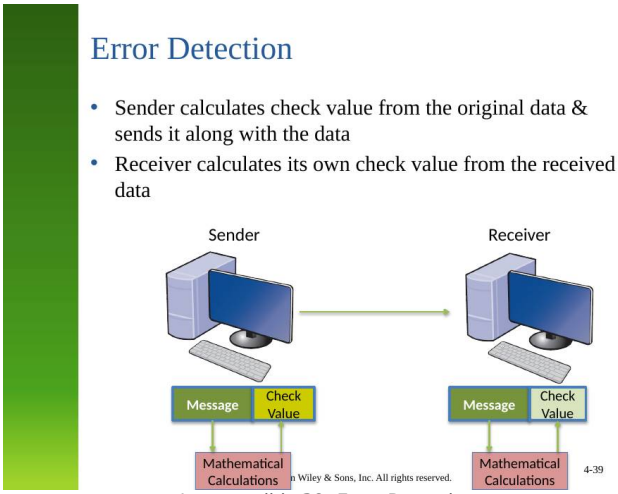
↳ *Same impairments as Part F.2; thunderstorm failure of copper (impulse noise, lightning) and the fibre fix is Exam 2024 Q1.2 ([D1]).*

E.7 Error detection: parity, checksum, CRC (slides 38 to 46)

Idea: the sender adds a **check value** (from a mathematical formula) to the message; the receiver recomputes it from the received data; **match = assumed error-free; no match = error detected.**



Lecture slide 38: Error Detection



Lecture slide 39: Error Detection

1. Parity check. One extra bit makes the total number of 1s even (**even parity**) or odd (**odd parity**). Slide example: data **10110011** has five 1s (odd); with **even parity** add parity bit **1** → transmitted **101100111**. The receiver recounts. **Benefits:** simple, fast, low overhead. **Drawbacks:** detects only **single-bit** errors; misses many multi-bit errors (any even number of flipped bits goes unnoticed).

Data (7 bits)	Number of 1s	Even-parity bit	Sent
1011001	4	0	10110010
1011011	5	1	10110111
0000000	0	0	00000000
1111111	7	1	11111111

2. Checksum. Split the data into blocks, add them. Slide example: 10110011 01010101 → blocks **179** and **85** → **264**. (Real TCP/UDP/IP use **1's-complement** addition.) The receiver does the same sum; equal = accepted.

3. CRC (cyclic redundancy check). Binary **polynomial division** (modulo-2, XOR) gives a remainder appended to the data. Detects **single-bit, double-bit and burst errors**; much more reliable than parity or checksum. Used in **Ethernet, Wi-Fi, USB, Bluetooth, CDs/DVDs, ZIP and PNG**. Slide example: data **1101011011**, generator **10011** (both sides agree on it beforehand). Sender appends zeros (one fewer than the generator length), divides, gets the remainder, appends it:

Step	Value
Data	1101011011
Generator	10011
Data + zeros	11010110110000
Remainder (CRC) from modulo-2 division	**1110**
Transmitted frame (data + CRC)	11010110111110
Receiver divides the frame again	remainder **0000** = no error detected

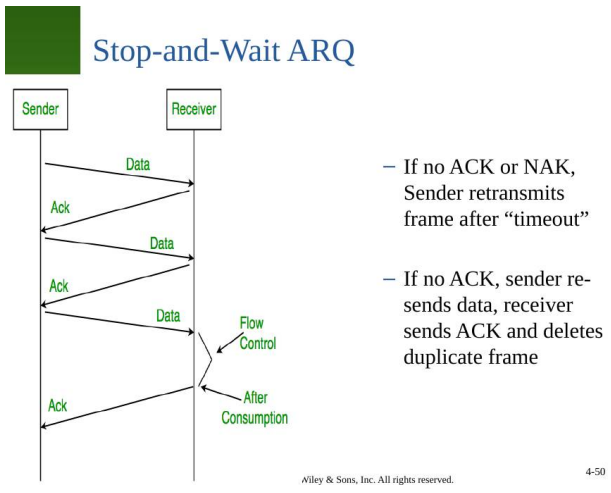
Note: the lecture slide quotes a remainder of 1010 for this example, but the modulo-2 division actually gives 1110 (check: dividing the slide's frame 11010110111010 by 10011 leaves a non-zero remainder, whereas data + 1110 divides exactly). The method is what the exam marks: show the division, and quote the remainder you derive.

Hamming code (exam scope, not in the slides). Parity only **detects**. **Hamming code** adds several parity bits at **positions that are powers of two (1, 2, 4...)**, each checking a different overlapping group of bits. If one bit flips, the pattern of failed checks (the **syndrome**) spells out **the position of the bad bit**, so the receiver can **correct it**. Example, data 1011 → codeword **0110011** (p1 p2 d1 p3 d2 d3 d4). If position 5 flips, checks 1 and 3 fail and 2 passes: binary 101 = **5**, so flip bit 5 back.

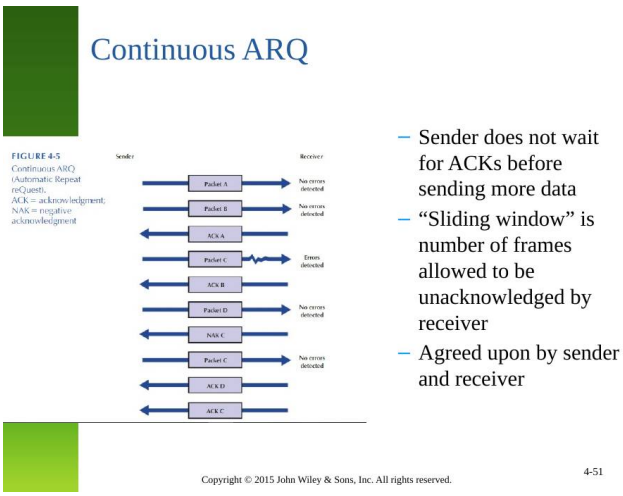
[T1] 2024 Q1.7 (6 marks): detection vs correction. Error detection (parity) only tells you **that** an error occurred (receiver must ask for a resend). **Error correction (Hamming)** uses extra redundant bits to find **which bit** is wrong and **fix it without retransmission**, at the cost of more overhead.

E.8 Error correction: ARQ and FEC (slides 47 to 52)

Approach	How it works	Notes
Backward error correction (retransmission)	When the receiver detects an error, it asks the sender to **send again**	Simplest and most common
ARQ (Automatic Repeat reQuest)	Manages retransmission with **ACKs, NAKs and timeouts**	Types below
Stop-and-wait ARQ (slide 50)	Send one frame, **wait** for ACK; if no ACK/NAK before the **timeout**, resend; the receiver **deletes duplicates**	Simple, slow
Continuous (sliding-window) ARQ (slide 51)	Sender **does not wait for each ACK**; the **window** = number of unacknowledged frames allowed, agreed by both ends	Basis of TCP flow control
Forward Error Correction (FEC) (slide 52)	**Redundant bits** let the receiver **detect and correct** without asking again	For satellites, live/video streaming, VoIP, where retransmission is **impractical, expensive or too slow**



Lecture slide 50: Stop-and-Wait ARQ



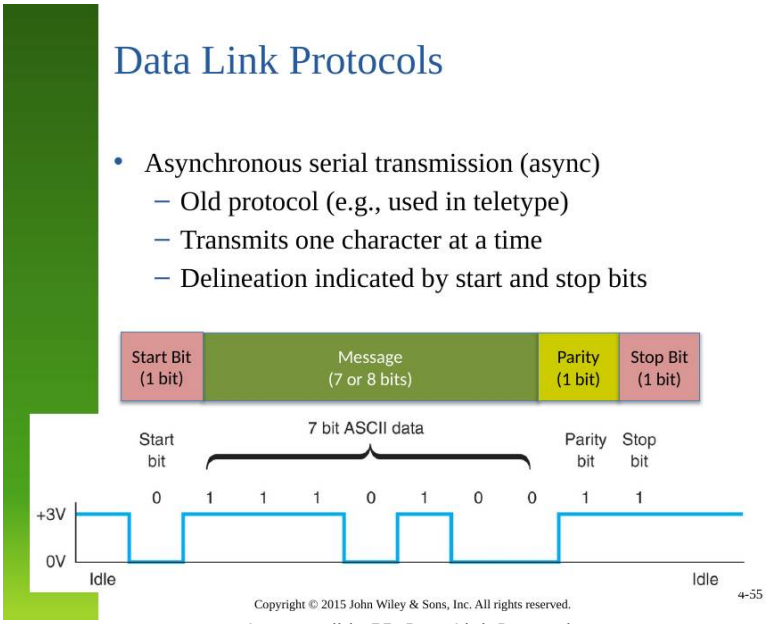
Lecture slide 51: Continuous ARQ

↳ *ARQ appears again in TCP (Part D.6) and on the wireless link (Part D.7). FEC reappears in streaming (Part C.5). Slide 53 says the following protocol slides are **self-study**.*

E.9 Data link protocols and message delineation (slides 54 to 59)

How does the receiver know where a message starts and ends? Protocols are classified as **asynchronous** or **synchronous**, and differ in **message delineation, frame length and field structure**.

Protocol	Key points
Asynchronous serial (old, teletype)	One **character at a time**; **start and stop bits** mark the boundaries; plus 7 or 8 data bits and a parity bit
Synchronous	Data in a large **frame** with **addressing** and **synchronisation characters**; examples **SDLC, HDLC, Ethernet, PPP**
SDLC (IBM)	Bit-oriented; flag, address (8 bits), control (8), data, **CRC-32**, flag; **bit stuffing (zero insertion)** solves the transparency problem
Ethernet II / IEEE 802.3	Most widely used LAN protocol; contention access. Fields: preamble and delimiter (8 bytes), destination address (6), source address (6), type (2), data (**46 to 1500 bytes**), CRC (4), interframe gap (12): **38 bytes of overhead** in total
PPP	Common **WAN** protocol: flag, address (1), control (1), protocol (2), data, CRC (2 or 4), flag



Lecture slide 55: Data Link Protocols

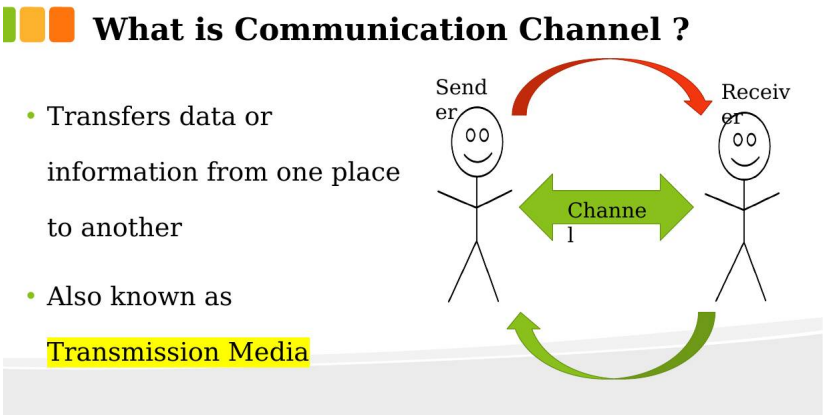
↳ *The Ethernet frame's 6-byte addresses are the MAC addresses of Part B.2; the 4-byte CRC is E.7.*

PART F: The Physical Layer and Communication Channels

Source: Communication Channels deck (52 slides) and Physical Layer deck (29 slides, one hidden). Textbooks: Tanenbaum Ch. 2, Forouzan Ch. 3 to 6. **Exam tags:** [T1], [D1].

F.1 What a communication channel is (Communication Channels, slides 2 and 3)

Every WhatsApp message or YouTube video travels through a **communication channel**: the path or medium that carries information from **sender** to **receiver** (also called **transmission media**). Three parts: **sender → channel → receiver**. Channels can be **physical** (cables) or **wireless** (radio waves, satellites).



Lecture slide 3: What is Communication Channel ?

↳ *Layer 1 moves raw bits; Layer 2 (Part E) groups them into frames. Everything that goes wrong here (noise, attenuation) shows up as errors in Part E.*

F.2 Transmission impairment (slides 4 to 10)

In an ideal world data arrives unchanged. In reality signals weaken or distort: **transmission impairment**, causing slow speeds, poor calls, video buffering and communication failure. Three causes:

Impairment	Meaning	Everyday example	Fix
Attenuation	**Gradual loss of signal strength** with distance ("the farther it travels, the weaker it becomes")	Your voice at 5 m vs 500 m	**Repeaters** and **amplifiers** strengthen the signal before it is too weak
Distortion	Parts of the signal **arrive at different times**, so the signal changes shape	"computer" arrives as "comptuer"	Better-quality media, shorter distances
Noise	**Unwanted electrical/electromagnetic signal** mixing with the real one	Radio static, lightning, power lines, **EMI**	Shielding, moving or shielding wires

Distortion causes: different frequencies travel at different speeds, poor-quality media, long distances. Effects of all three: data errors, lost packets, reduced performance, poor audio/video quality.

↳ *[T1] Layer 1 "attenuation / weak signal" (Computer B, 2026 Q1) and EMI; [D1] "attenuation, EMI". The same three explain bit errors in wireless (Part A intro).*

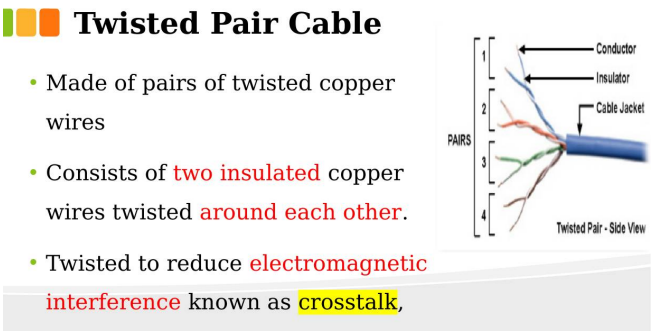
F.3 Two families of media (slides 11 to 15)

	Guided (wired)	Unguided (wireless)
Signal travels	Through a **physical cable**	Through the **air/free space** as **electromagnetic waves**
Examples	Twisted pair, coaxial, fibre optic	Radio waves, microwaves, infrared, satellite
Characteristics	Faster, more reliable, less interference, **more secure**, limited by cable length	Supports **mobility**, quick to install, short or very long range, **more susceptible to interference**, obstacles and weather

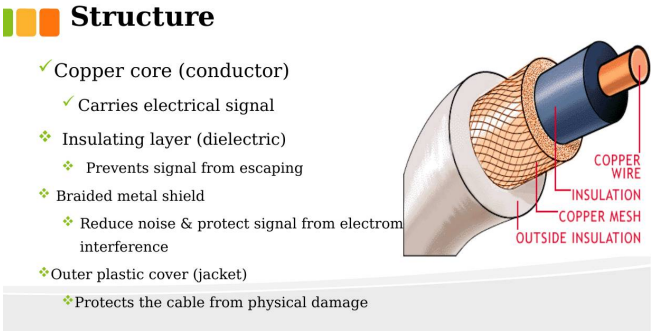
Why do businesses wire their servers? **Faster, more reliable, more secure, lower latency, less interference.** Modern networks **combine** both: your phone uses Wi-Fi (unguided) to a router connected to the

internet by fibre (guided).

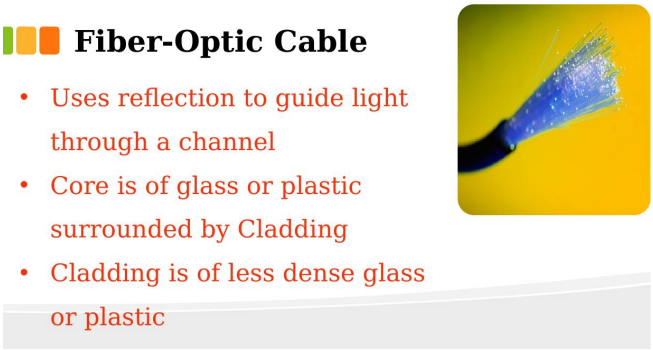
F.4 Guided media (slides 16 to 29)



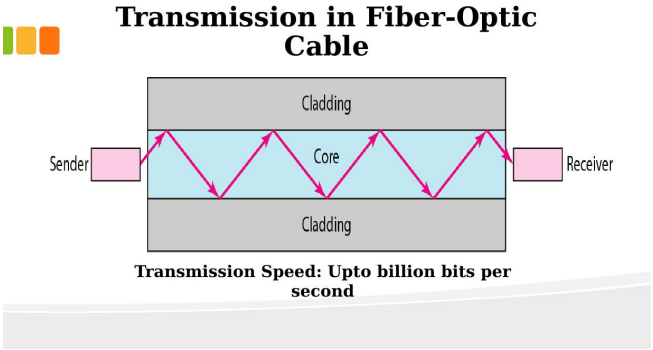
Lecture slide 16: Twisted Pair Cable



Lecture slide 20: Structure



Lecture slide 25: Fiber-Optic Cable

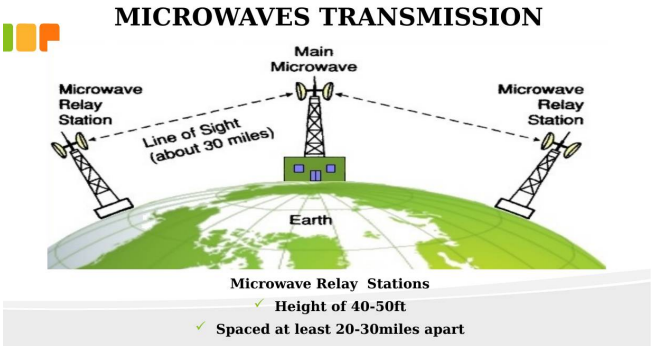


Lecture slide 26: Transmission in Fiber-Optic Cable

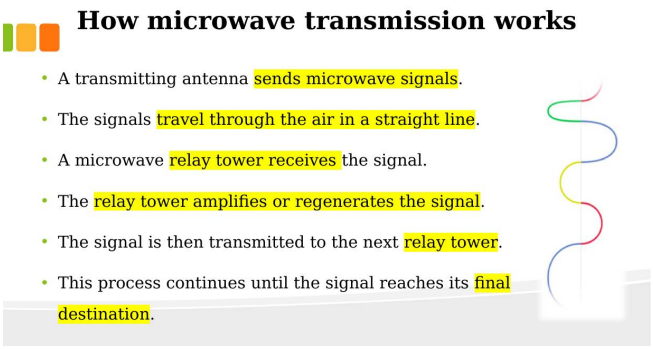
	Twisted pair	Coaxial cable	Fibre optic
Structure	Pairs of insulated copper wires **twisted** around each other (twisting reduces **crosstalk**/EMI)	**Copper core**, insulating layer (dielectric), **braided metal shield**, plastic jacket	Very thin glass/plastic strands (**core + cladding**); **light pulses** guided by **reflection**
Signal	Electrical	Electrical	**Light**
Speed/ bandwidth	About 1 Mbps over short distances; lowest	Bandwidth about **80×** twisted pair; modern systems far higher (DOCSIS)	**Up to billions of bits per second**; highest
Uses	LANs, telephone landlines	TV, **cable internet**, CCTV, some computer networks	Fibre to the home, **telephone backbone**, data centres, banking, universities, cloud
Advantages	Cheap, easy to install, flexible, widely available	Higher bandwidth, better interference protection, reliable over longer distances, durable, multiple signals	**Greater capacity**, smaller and lighter, **lower attenuation** (travels far before amplification), **immune to EMI/lightning**
Drawbacks	Short distance, lower bandwidth, **susceptible to EMI**, slower	More expensive, thicker, harder to install, slower than fibre	More **expensive to install**, **fragile**, needs specialist equipment and skilled technicians, repairs harder

↳ *[T1] Q2 and Exam 2024 Q1.2: copper (twisted pair/coax) suffers **attenuation, EMI and lightning**, which is why **fibre** fixes the Sekhukhune thunderstorm failures.*

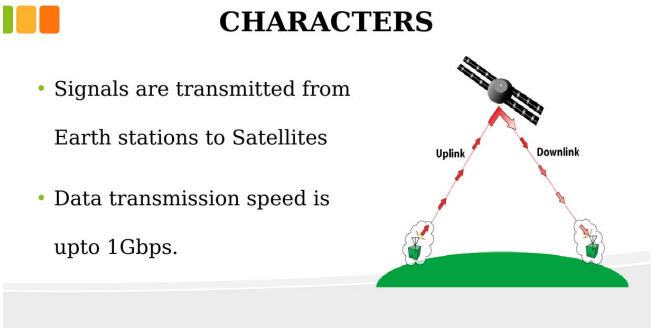
F.5 Unguided media (slides 30 to 50)



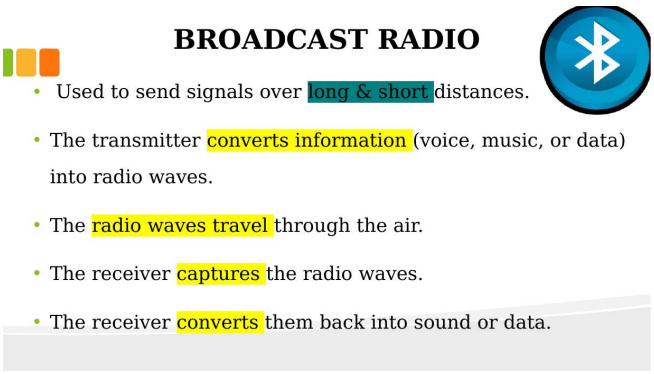
Lecture slide 33: MICROWAVES TRANSMISSION



Lecture slide 34: How microwave transmission works



Lecture slide 41: CHARACTERS



Lecture slide 45: BROADCAST RADIO

Microwaves (33 to 37): line-of-sight: a clear straight path is needed, and they cannot follow the Earth's curvature, so **relay towers** (about 40 to 50 ft high, **20 to 30 miles apart**) receive, amplify and forward the signal. **Unidirectional**, point-to-point, about **150 Mbps**; high-frequency microwaves cannot penetrate walls. Benefits: high speed, long distance, **no cable across difficult terrain**, often faster and cheaper to deploy than fibre in remote areas. Drawbacks: **needs line of sight** (mountains, tall buildings, trees block it), **rain/storms** reduce quality, limited distance per hop.

Communication satellites (38 to 43): an orbiting satellite **receives, amplifies and retransmits**: Earth station → **uplink** → satellite → **downlink** → another Earth station (like a person on a mountain relaying a message). Up to about **1 Gbps**; carries huge amounts of data, covers very large areas, connects remote places, enables broadcasting. Drawbacks: affected by bad weather, security problems, **costly** to launch and maintain, long distance means a small **delay**.

Radio waves (44 to 49): the oldest and most-used wireless form; travel long distances and can pass through walls. **Broadcast radio:** transmitter converts information to radio waves, receiver converts back. A **transceiver** both sends and receives (phone, Bluetooth earbuds). **Bluetooth** about 1 Mbps. **Cellular radio:** the area is divided into **cells**, each with a tower (base station); as you move your phone switches towers: a **handover (handoff)**.

Infrared (50): line of sight, short range (for example a wireless keyboard or mouse).

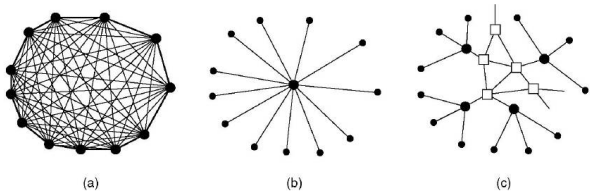
> **[T1] 2026 Q2 (10 marks): offices 25 km apart across tall buildings and mountains.** (a) **Microwave is unsuitable because it needs an unobstructed line of sight**; buildings and mountains block it. (b) **Fibre optic** is best: very high capacity (bandwidth), low attenuation over 25 km, immune to EMI/lightning (also secure). (c) If mountains prevent laying cable: use **microwave relay towers on high ground** (or hilltops/masts) so each hop has line of sight, repeating the signal between them.

↳ Handover from the cellular-radio slide is the handoff of Part A.13 ([T2] 2026 Q3.1). Unguided media's interference problems lead to CSMA/CA in Part E.4.

F.6 The telephone system and multiplexing (Physical Layer deck, slides 2 to 13)

Structure (slides 2 to 4): connecting every phone to every other (full mesh) is **extremely expensive and impractical**; one **central switch** is better; a **two-level hierarchy** (local switches connected by higher-level switches) is the foundation of the modern system. Components: **local loops** (analog twisted pairs to homes), **trunks** (digital fibre between switching offices), **switching offices** (where calls move from one trunk to another; "the brain").

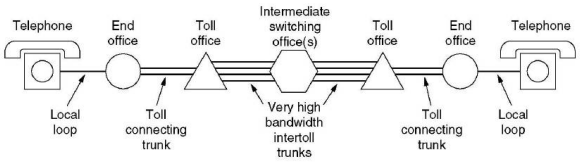
Structure of the Telephone System



(a) Fully-interconnected network.
(b) Centralized switch.
(c) Two-level hierarchy.

Lecture slide 2: Structure of the Telephone System

Structure of the Telephone System (2)



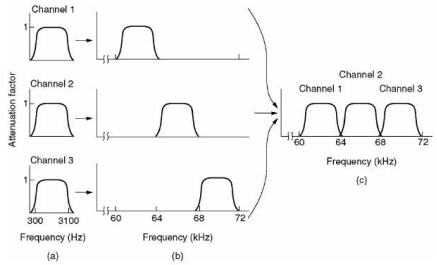
A typical circuit route for a medium-distance call.

Lecture slide 3: Structure of the Telephone System (2)

Multiplexing combines several signals into **one signal over a shared medium** to share a scarce resource: one trunk can carry thousands of calls; without it every call would need its own wire.

Technique	How	Medium
FDM (Frequency Division)	The channel's bandwidth is divided into **frequency bands**; each signal is **shifted to its own band**, combined, then separated by **filters** at the receiver	Copper/coaxial
WDM (Wavelength Division)	The **optical version of FDM**: each channel uses a different **wavelength (colour) of laser light**; hundreds of channels per fibre, 10 Gbps+ each, **terabits per second** per fibre	**Fibre**
TDM (Time Division)	A high-rate channel is cut into **fixed time slots** repeating in a cycle; each signal gets one or more slots; demultiplexed at the receiver	Digital

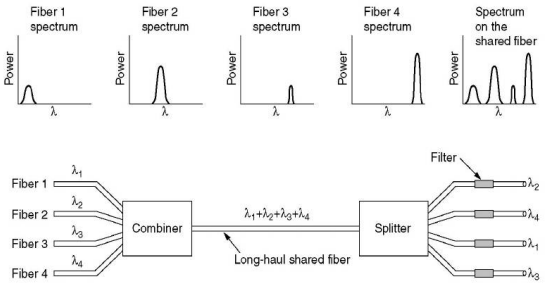
Frequency Division Multiplexing



(a) The original bandwidths.
(b) The bandwidths raised in frequency.
(b) The multiplexed channel.

Lecture slide 7: Frequency Division Multiplexing

Wavelength Division Multiplexing



Wavelength division multiplexing.

Lecture slide 9: Wavelength Division Multiplexing

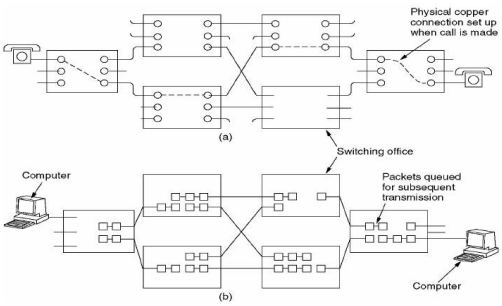
TDM example (slide 13): four employees share one line; each gets a time slot, only that employee's voice is sent during it, and the cycle repeats thousands of times a second so everyone seems to talk continuously.

> ↳ *Slotted ALOHA's time slots (Part E.5) and reservation MAC (E.2) use the same idea as TDM. [D1] lists **QAM, WDM, BER**; the 2024 Q1.4 question on a 200 km fibre link with WDM and QAM is in F.8.*

F.7 Switching techniques (slides 14 to 29)

Switching = selecting the path data takes from the available options.

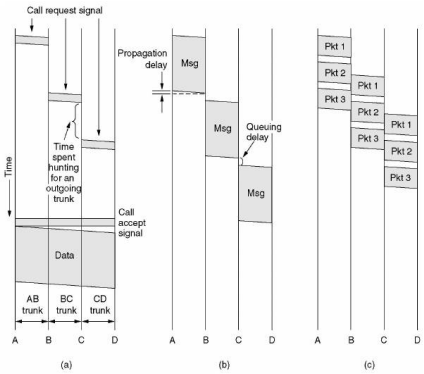
Circuit Switching



(a) Circuit switching.
(b) Packet switching.

Lecture slide 14: Circuit Switching

Switching techniques



(a) Circuit switching (b) Message switching (c) Packet switching

Lecture slide 15: Switching techniques

	Circuit switching	Message switching	Packet switching
Idea	A **dedicated path** is set up **before** data is sent and **reserved** until the session ends	The **entire message** is one unit, **stored completely** at each node, then **forwarded**	The message is cut into **packets**, each with source/destination address and **sequence number**, sent **independently**
Phases	**Setup** (request, find and reserve path, accept) → **data transfer** (continuous, no routing decisions) → **termination** (release)	Hop by hop: receive and store whole message, check errors, find best route, forward when the link is free	Packets may take **different routes**, **stored and forwarded** at each router, **reassembled** at the destination; lost/corrupt ones are retransmitted
Delay	**Predictable**, good for **real time voice**	**High** (store and forward the whole message); unsuitable for voice/video	Variable but low overall
Efficiency	**Wasteful**: capacity unused when idle, nobody else can use it	Efficient channel use; no need for both ends to be available at once; multi-destination; supports different speeds; reliable (re-routing)	**Highly efficient** use of bandwidth; robust (reroutes around failures); good for **bursty** traffic (Internet); many users share a link
Drawbacks	Sender and receiver must both be available; **long setup (about 10 s)**; **more expensive**; dedicated path per connection	**Not for real-time or interactive traffic**; nodes need **large storage** because message length is unlimited	Out-of-order arrival, reassembly overhead
Best for	Phone calls	Overnight bulk/e-mail-like transfers	The Internet

[T1] 2026 Q3 (10 marks): hospital with real-time voice and overnight backups. (a) **Voice → circuit switching** (some answers say packet switching with QoS): dedicated path, **predictable/constant low delay**, no reordering. (b) **Backups → packet switching** (or message switching): no hurry, efficient bandwidth sharing, can reroute around failures, delay does not matter. (c) **Technician trap, "message switching for voice because nodes store and forward the complete message": wrong**: store-and-forward of a whole message adds long, unpredictable delay, which is exactly what real-time voice cannot tolerate; message switching suits delay-tolerant transfers.

↳ *Packets and sequence numbers connect to TCP reassembly (Part D), IP fragmentation (B.9) and the Internet being packet switched. [T1] "Circuit vs Message vs Packet Switching".*

F.8 Exam extras for [D1] (not in these slides; verify against your notes)

Topic	Short answer
Throughput (2016 Q1.1)	The **actual** rate of successful data delivery. Six commonly cited factors: **bandwidth/capacity, signal quality/errors (BER), delay/latency, congestion, protocol overhead, and hardware/device limits**.
Simplex / half-duplex / full-duplex (Exam 2024 Q1.3)	**Simplex**: one direction only (a public address system, radio broadcast). **Half-duplex**: both directions but **one at a time** (walkie-talkie, hub Ethernet, Wi-Fi radios). **Full-duplex**: both directions **at once** (telephone, switched Ethernet).
QAM	Quadrature Amplitude Modulation: varies **amplitude and phase together** to send several bits per symbol; higher order = more speed but **less noise tolerance**.
High BER on a 200 km fibre link with WDM/QAM (Test 2 2024 Q1.4)	**Causes:** attenuation over distance, dispersion (distortion), crosstalk between wavelengths, amplifier noise, a modulation order too high for the signal quality. **Fixes:** add optical amplifiers/repeaters, dispersion compensation, wider channel spacing, **lower-order QAM**, **FEC** (Part E.8).
BER	Fraction of received bits in error; it is time-varying on wireless (Part A intro).

↳ *Throughput connects to Part D.8 (measuring performance) and the $(1 - p)^n$ frame-survival idea (error rate × frame size) from the wireless part.*

PART G: One-page logic map and final checklist

G.1 How the parts connect

Question a network must answer	Layer	Where to read
How do bits travel and what damages them?	Physical	F.2, F.4, F.5
How are many signals / calls shared on one medium?	Physical	F.6, F.7
How do frames get delivered, checked and fixed?	Data link	E.6 to E.8
Who may transmit on a shared channel?	Data link (MAC)	E.2 to E.5
How do we address, group and route across networks?	Network	B.2 to B.12
How does a host keep its connection when it moves?	Network	A.1 to A.6
How is end-to-end delivery made reliable?	Transport	D.2 to D.7
How do applications find servers and use the network?	Application	C.0 to C.6
How is it all protected?	Security (all layers)	A.7 to A.12

G.2 Pre-exam checklist (tick each one)

Skill	Done?
Manual VLSM table (descending order, first/last/broadcast, next free address)	☐
Spot the "technician trap" (address in the wrong range; broadcast given to a host)	☐
TCP three-way handshake with non-zero ISNs, plus the diagram	☐
Match services to ports (80, 443, 22, 25, 53, 21)	☐
Hidden terminal problem + RTS/CTS + why not CSMA/CD on Wi-Fi	☐
Direct vs indirect Mobile IP (4 steps each + latency + transparency)	☐
Asymmetric encryption (recipient's public key encrypts)	☐
DoS: attack, CIA principle (Availability), two controls	☐
Parity vs Hamming; CRC idea; ARQ vs FEC	☐
Circuit vs message vs packet switching and the voice trap	☐
Fibre vs copper vs microwave (line of sight) selection	☐
DNS resolution steps and DHCP DORA	☐